SAINT MICHAEL COLLEGE OF CARAGA

LIBRARY MODERNIZATION: INFLUENCE ON STUDENTS' LIBRARY EXPERIENCE IN SELECTED ACADEMIC LIBRARY IN AGUSAN DEL NORTE

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF LIBRARY AND INFORMATION SCIENCE

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NOVEMBER 2026

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PUBLISHABLE JOURNAL 100

SAINT MICHAEL COLLEGE OF CARAGA

SAINT MICHAEL COLLEGE OF CARAGA
College Of Computing and Information Sciences

CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"**Library Modernization: Influence on Students' Library Experience in Selected Academic Libraries in Agusan Del Norte**"** presented and submitted by Angel Mica J. Gonzales, Ashlly J. Llavan, Cherrish Jane T. Ranario, Renalyn M. Beniga, and Kevin Dominic M. Suan. In partial fulfillment of the requirements for the degree of Bachelor of Library and Information Science is hereby accepted and recommended for Oral Examination.

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ABSTRACT

TITLE: LIBRARY MODERNIZATION: INFLUENCE IN STUDENTS' LIBRARY EXPERIENCE IN SELECTED ACADEMIC LIBRARIES IN AGUSAN DEL NORTE

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ADVISER: Lani Jean E. Lumogda, RL
PLACE OF PUBLICATION: Nasipit, Agusan del Norte
DATE 00/00/0000
PAGES 00 Pages

(RESEARCH DESIGN)

I. OBJECTIVES

The objective of this study is to assess the influence of modernized libraries on the experiences of students, evaluating as well as the resources and the environment itself, particularly along the academic institutions of Agusan del Norte. Moreover, this aims to evaluate the impact of library modernization on the library experience of students with the objective of scrutinizing key areas for engagement, as well as to support efforts in improving library services. Additionally, the study seeks to explore how modernization efforts in academic libraries affect

the experiences, satisfaction, and adaptability of student users. By recognizing the key factors that enhance or hinder the experiences of students in modernized libraries, the finding of the study can suggest and impose development on more effective library systems for educational institutions and policymakers.

II. METHODOLOGY

This study will utilize a descriptive-correlational research design within a quantitative approach. Information will be gathered on how academic libraries adapt to modernization to enhance the quality of services, improve accessibility of resources, and meet the evolving needs of students in the digital age. The research will focus on 3 academic libraries within Agusan del Norte including both private and public institutions, with a total of 150 students, representing various courses and year levels, comprising 50 respondents each. Data will be gathered through paper-and-pencil survey questionnaires ensuring high response rates, accessibility and immediate clarification of participant inquiries, and better control over the data-gathering process. The instrument will use a Likert scale format to measure the degree of satisfaction of the students within the library. Data will be analyzed using descriptive statistics to summarize the level of library modernization and the overall experiences of the students, and inferential statistics such as Pearson's correlation to determine the relationship between library modernization and student satisfaction and experiences, and multiple regression analysis to identify which modernization factors significantly influence students' adaptability and engagement in the library setting.

However, before dissemination, the survey questionnaire will undergo a validation and reliability testing process to ensure that it effectively measures the intended variables. The

instrument was adopted and modified from an existing and validated study of Lei Peng et al. (2022) and Bawasanta et al. (2025)

Prior to data gathering, each participating institution will be asked for permission to conduct the study. The researchers will personally visit the academic libraries in Agusan del Norte to administer the survey questionnaires to students who will be present and available during the conduct of the study. Each respondent will be provided with a brief explanation of the study's purpose, assurance of confidentiality, and informed consent before completing the questionnaire.

In addition, a baseline survey—the Library Information Form—will be administered to the participating head librarians of academic libraries within Agusan del Norte to gather general information about the modernization level of their libraries. The information will serve as the basis for confirming eligibility as a research locale.

III. FINDINGS

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IV. RECOMMENDATIONS

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KEYWORDS: Library Modernization, Students' Library Experience, Academic Libraries

DEDICATION

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Worldwide technology and digitalization are changing at a breakneck pace, and these changes are impacting the way we communicate, educate, and get information (Trecene, 2021). Academic libraries must adapt to these developments. According to Carlvalho and Mandrekar (2021), students' success in school depends on how often they use the library. Libraries are instrumental in a student's learning, studying, and personal development (Jimenez et al., 2024). Historically, libraries were simply places to get books, but now they have become student-friendly centers with a variety of services and facilities.

Libraries today include digital tools like computers, e-books, and online resources. With the use of technology in students' studies, the demand for digital materials has been continuously increasing, as stated by Galand and Villa (2024). Dorado (2019) also noted that this trend has been the primary factor necessitating libraries to change their modes of operation. Per the research of Leonido et al. (2024), a library is a part of the school that needs to be upgraded, and the staff should be trained to be technologically skilled. Likewise, numerous universities worldwide are updating and revitalizing their libraries (Banleman et al., 2023). One example is Baltimore, USA, where Diller and Wallin (2023) reported that schools' libraries are being remodeled to facilitate students' learning and make them feel more comfortable. The evidence for this was also the finding of the research conducted by Mehta and Cox (2019). Students in the study said that academic libraries made them feel "at home". Although the changeover to the latest technology has its upsides, it also has downsides. In Indonesia, Winata et al. (2020) found that many students struggled to adapt to the new library system after the COVID-19 pandemic.

In India, Burhansab et al. (2020) stated that modern libraries have more advanced systems but still face challenges, such as staff shortages and maintenance issues. Kuyela et al. (2025) also observed that the shift from physical to digital services has affected libraries in developing countries as they seek to meet students' growing needs.

Such endeavors are also taking place in the Philippines. As per Abrigo and Labangon (2021), many academic libraries are investing in upgrading their services and resources. Filipino librarians were reported by Fernandez et al. (2025) to be tech-savvy and service-oriented, while Corpuz (2020) noted that there is still significant room to enhance electronic resources. Tuble and Bayoneta (2019) also said that success in library modernization depends on local actions and resources. Nevertheless, only a handful of articles have been published that discuss how the library upgrade affects students' learning and their ability to adapt to new systems. According to the research by Abamhan and Armilla (2025), the lack of technological support and the poor condition of infrastructure in some schools have prevented these schools from modernizing quickly (Dorado, 2019). The research by Cabonero et al. (2025) also indicates that a large number of students lack a regular, convenient way to access the Online Public Access Catalog, which, in turn, hampers their learning and makes them less satisfied. Zhu et al. (2025) noted in their study that the library is slowly falling off the track to be called the "heart of the university" as the number of students going there drops. Lascano (2021) added that many students have little knowledge or interest in Library and Information Science.

This is why the current research seeks to determine how library modernization influences students' satisfaction, experience, and adaptability, in line with the research by Mehta and Cox (2019). According to Lasig (2024), libraries should never stop updating since students' needs are constantly changing. Moreover, Guilles (2019) said that if librarians have a clear picture

of the libraries' current condition, they can prepare more efficient plans for future improvements.

Libraries in rural areas of Agusan del Norte, a province in Mindanao, still use age-old systems and face resource constraints, hindering students' access and experience (Maraon & Agcito, 2024). This study pledges to reveal the impact of library modernization on students' experiences in the academic libraries of Agusan del Norte. It is committed to identifying the aspects that need improvement and providing the necessary resources to those who will create new initiatives to enhance library services. The study employs the Information Systems Success Model (ISSM) of Delone and McLean (1992; updated 2003), Technology Acceptance Model (TAM) of Davis (1986), and User Experience (UX) Theory by Norman and Nielsen (1995) to examine how modernization influences students' satisfaction and their views of library facilities. The main goal of this study is to assess how modernization efforts in libraries affect students' satisfaction and experiences in universities in Agusan del Norte.

By identifying what helps or hinders students' use of modernized libraries, the study aims to offer ways to improve school library systems and inform policymakers. Students can benefit from better facilities and services that support their studies, librarians can learn how to provide more effective services, and schools can use the findings to improve their modernization plans. The goal of this research is to help libraries in Agusan del Norte enhance their systems and provide better learning spaces for students.

1.2 Review of Related Literature

This chapter presents a review of the literature relevant to the study. It gives background information to help readers understand the research. The reviewed studies demonstrate how this research relates to library modernization and students' library experience. To know how library modernization affects student satisfaction, the study examines several factors. These include the digital infrastructure,availability of online resources, automation of library services, and enhanced physical learning spaces. It also considers students' backgrounds or profiles. The study also examines the ease of access to library service, perceived usefulness of modernized features, user satisfaction, and perceived impact on academic productivity. This part explains how these factors affect students' use of modern libraries and their satisfaction with services.

Academic Libraries

Academic libraries serve as foundational institutions, being instrumental to student development, learning, and academic success (Jimenez et al., 2024). Historically viewed as mere repositories for books, these facilities have fundamentally transformed into comprehensive, student-friendly centers offering a variety of facilities and resources. Research consistently demonstrates that a student's level of success in their studies is directly correlated with the frequency of their library utilization (Carlvalho and Mandrekar, 2021). However, the effectiveness of modernization is intrinsically tied to the academic institution's capacity. Difficulties in achieving instant modernization can arise when schools have limited infrastructural support and only moderate technological resources (Abamhan and Armilla, 2025; Dorado, 2019). Moreover, a school's dedication to valuing and supporting its library system significantly influences how students perceive and engage with it (Sumartin, 2021). While Filipino librarians are often recognized for their service-oriented nature and technological proficiency (Fernandez et al.,

2025), there remains a critical need to continuously enhance the quality of electronic resources offered (Corpuz, 2020). If the trend of declining student visits continues, the library risks diminishing its status as the "heart of the university" (Zhu et al., 2025).

Library Modernization

The global acceleration of technology and digitalization necessitates continuous and rapid adaptation for academic libraries, which is redefining how students engage in learning and seek information. Modernization signifies a crucial shift from traditional, book-centric facilities to incorporating digital tools such as e-books, online databases, and computing resources. This integration is vital because the increasing use of technology in academic work has fueled a sustained demand for digital learning materials (Galand and Villa, 2024). While the push for modern systems is widespread, it presents unique challenges. For example, some students faced adaptation struggles when new library systems were implemented post-pandemic (Winata et al., 2020) , while other modernized facilities still contend with fundamental issues like staff shortages and resource maintenance (Burhansab et al., 2020). Furthermore, the transition from purely physical to digital service models is a particularly complex undertaking for libraries in developing nations as they strive to meet the evolving requirements of their student populations (Kuyela et al., 2025). In the Philippine context, many academic libraries are actively addressing this need by upgrading their services and resource infrastructure (Abrigo and Labangon, 2021).

Students' Library Experience

The effectiveness of any modernization effort is primarily gauged by its influence on the Students' Library Experience, which directly leads to student satisfaction, engagement, and successful adaptation to new systems. This positive experience is cultivated by a confluence of

factors, including the library's physical attributes such as its overall design, available facilities, and spatial variety. Specifically, user satisfaction is significantly increased by ensuring the accessibility and high quality of services and facilities (Tulung et al., 2024), which now commonly include features like internet access, digital catalogs, and collaborative study areas. Furthermore, the aesthetic elements of the interior design—such as effective lighting, cohesive color schemes, and ergonomic furniture layout—are critical, as they foster a greater sense of comfort and encourage deeper user engagement (Aabouelala, 2022). A well-managed physical environment, particularly the provision of quiet and distraction-free spaces, is essential for promoting focus and ultimately motivating students to excel in their studies (Scoulas & De Groote, 2022). Conversely, barriers to access, like the unavailability of basic tools such as the Online Public Access Catalog (OPAC), have been shown to directly result in user disappointment and diminished learning outcomes (Cabonero et al., 2025). Therefore, research in this area is paramount for isolating the specific elements that either enhance or hinder the student experience, enabling institutions to strategically improve their systems and maximize student satisfaction.

Digital Infrastructure

The transformation of the library into a dynamic information hub through the strategic investment in a robust digital infrastructure has made a fundamental change in the modern academic libraries. The Fourth Industrial Revolution (4IR), which integrates both digital and physical technologies, has rendered this change unavoidable. For this reason, academic libraries will be compelled to synchronize their systems in such a way that they will be globally relevant (Moonasar, 2024, p. 235; Vaish, 2024).

At the base of the modernization, there is the establishment of a Digital Library that actively promotes the generation of knowledge and transforms libraries into proactive facilitators instead of keeping them as passive centers (Acta Informatica Malaysia, 2023). The term Digital Library refers to a formal collection of data that is digitally stored for retrieval through a network and is supported by various services (Upadhayay, 2020). A variety of technology components support this infrastructure, including the most advanced ones like artificial intelligence (AI), machine learning, and big data analytics, which are the main factors of the tremendous improvement of service accessibility, efficiency, and user experience.

Moreover, this foundation consists of high-end technologies such as cloud computing, RFID, blockchain technology, Internet of Things (IoT), and Integrated Library Management Systems (ILMS) (Ibrahim, 2024). The digital backbone relies on a functional computer network for the communication of vital electronic resources (Amosi, 2020). Besides, these resources make the reliable and current information accessible to students and researchers as additional benefits (Mahanta, 2019). This change has forced the librarians to take up a digital hybrid service delivery model and to rethink their professional roles (Kabatangare, 2019; Ibrahim, 2024).

Availability of Resources

E-resources, or electronic resources, have been making quite an impact in the academic world at the beginning of the twenty-first century, which has been a very radical change in the knowledge-seeking behavior of students and researchers globally (Tofi and Fanafa, 2019). Consequently, e-resources have evolved into an indispensable source of knowledge in higher education, allowing both undergraduate and graduate students to get the most updated and relevant information from a wide range of topics. The emergence of e-resources has brought

about a major transformation in the way information is handled and managed in academic institutions (Goodluck Ifijeh & Felicia Yusuf, 2020).

For a library to successfully integrate these into its modernization efforts, Availability of Resources is crucial. Availability, as defined in the literature, refers to the necessary provision for and inclusion of these resources in the library's collection. Operationally, this translates to users having "readily access to information in electronic format with little or no stress" (Tofi and Fanafa, 2019). The successful availability of resources calls for two basic prerequisites apart from mere ownership: the users should not only know of the resources' existence but also possess the necessary skills to use them effectively. The electronic resources, when their availability is ensured and they are used properly, will always provide quick, accurate, reliable, and authoritative information. This will not only support but also increase the pace of innovation in education and research. This emphasis on resource delivery is integral to the broader transformation of academic library services from an outdated non-digital to a hybrid service model, meeting 21st-century user needs (Kabatangare, 2019; Goodluck Ifijeh & Felicia Yusuf, 2020).

Physical Learning Spaces

In addition to embracing technology, academic libraries are being transformed to embody the crucial function of the physical setting in affecting students’ involvement and academic success (Dianing & Acedera, 2025). To make their learning environments more unique and practical, the librarians have to constantly analyze the users' behavior and put the evolving needs first in the planning and design process (Somaratna, S. D., 2022).

Interior design is one of the main factors that not only defines but also influences the very setting of scientific inquiry and innovation. (Abouelela, 2022). It is through the application of better interior design that we can finally get rid of the old library image, as it provides open reading areas and many seating choices and types of furniture, light, inviting, and attractive spaces. The researchers came to the conclusion that the physical environment is very important as it shows that the relationship between space usage and total student learning pleasure is through contentment with learning space (Cui & Ahn, 2024).

Libraries are required to pass through their classic functions step by step and at the same time put into practice user-centered design methods, which give prominence to the characteristics of flexibility, comfort, and inclusivity (Cui, Liao, & Ahn, 2025; Cui & Ahn, 2024), to finally satisfy the various learning styles and encourage the development of autonomy, competence, and engagement. The repercussions of such a shift towards modernization are quite obvious: investing in ergonomic furniture and multifunctional learning areas should to a large extent be regarded as a necessity, since student happiness will be markedly increased, which in turn will lead to more active participation and better performance in academics (Cui & Ahn, 2024).

Ease of Access to Library Service

Ease of Access to Library Service is widely understood as the general skills of users to utilize the services effectively and efficiently without the strong obstacles. The idea is closely connected to the accessibility and ease of services, which include both physical and online aspects (Musa, Darman, and Yunus, 2023; Basiag, 2025).

Libraries in the digital era are supposed to become modern and tech-savvy facilities that can deliver easy access to a great number of information resources within the system, anytime and anywhere (Ashikuzzaman, 2025; Sayekti et al., 2022). This has been attained by a blend of improved services and customer-oriented systems. Some of the primary innovations are Digital Reference Services and mobile library apps, which are convenient and flexible and enable customers to access the library wherever they are (Iroroeavwo et al., 2024; Osisanwo and Adeola, 2024). Moreover, using OPAC and electronic collections is much more efficient and faster since they can access the information more quickly and easily when they are free, and on their own (Odunola and Tella, 2020; Kapadi and Patil, 2023).

Much to this convenience is the ease of access to the information system. Studies note that convenience and accessibility are the most important factors in establishing a positive connection with the library services (Sayekti et al., 2022; Rodenbarger and Hortsch, 2020). A poor-looking graphic interface or an incomprehensible menu on the library website may make the user lose their way, which is why it is necessary to keep the content in the search systematically organized and to have web navigation systems according to user behavior (Guchacha, 2019). Technology has removed geographical boundaries and enabled access to information in real-time and fast (Nakhumicha, 2024), however, at the same time, effective usage is also associated with knowledge and ease of navigating the offered material, which raises several obstacles, such as the absence of the required IT skills or non-cooperative employees (Ternenge & Kashimana, 2019; Xie and Babu, 2020). Finally, we need the provision of the accessibility of the relevant information to support teaching, learning, and research.

Perceived Usefulness of Modernized Features

Perceived Usefulness (PU) is a fundamental construct in technology adoption models and is described as an individual degree of the belief that the utilization of a particular system or technology will enhance the person's productivity or the value-addition to their life (Surairi, Adam, and Mukhlis, 2025). In fact, PU is the subjective likelihood of the user that the technology might aid in fulfilling their information requirements or enhance their work performance, e.g., performing a task in a shorter duration of time (Odunola and Tella, 2020; Innovation Acceptance Lab, 2020).

Measured in terms of modernized features, PU in the academic library setting is directly related to physical benefits. These functionalities, including the OPAC, are found valuable due to the ability to find electronic information resources fast and conserve time in finding materials, which influences student patronage greatly (Odunola and Tella, 2020). In addition to search systems, adoption of sophisticated, integrated technologies, such as IoT-based systems, such as automated retrieval, smart door lock, and self-checkout, are implemented in a specific manner to improve the security, material management, and provide a remarkable experience to users who demand quick and real-time services. These advantages, which coincide with the application of the Internet to make everyday activities easier (Kumar, 2023), confirm that the fact that the system can have a positive effect is a highly significant consideration that a person will make before making an ultimate choice to adopt or neglect the technology (Surairi, Adam and Mukhlis, 2025).

User Satisfaction

The concept of user satisfaction in an academic library is dynamic and multifaceted as it is associated with a favorable experience that the user gained after the interaction and their

likelihood of using the library in the future (Pradhan and Shrestha, 2024). The paradigm has changed its base, formerly centered on the real aspects of physical facilities, the professionalism of the staff, and the print collections (Umoh & Agwunobi, 2020). The fast adoption of mobile and computer technologies has contributed to the shift in the behavioral trends among users, which has led to the need to seek and rely more on e-resources and access to digital information than on traditional print media (Donovan, 2020).

The perceived quality of the service, especially digital services, in contemporary satisfaction is determined using four key dimensions including usefulness, functionality, reliability, and responsiveness (Suharto and Kadir, 2021). The level of satisfaction of a student is now holistically affected by a combination of elements that include the quality of library services, resources, staff capabilities, materials organization, physical environment, and the implementation of expert systems and technology (Alkali and Muhammed, 2024). Research proves that even when users are content with the physical infrastructure, they can simultaneously be dissatisfied with the library collection and computer facilities (Kaushamalika & Weerakoon, 2020), which suggests a root cause of the necessity to invest resources in balance. On the other hand, it also reports high satisfaction in terms of the supportive role of the library in the learning process and how it treats users professionally (Gyau et al., 2021). It is important to note the constant pursuit of high service quality, under which librarians offer active support and useful service (Barfi et al., 2023; Agina-Obu and Sawyerr-George, 2023).

The level of student satisfaction is a critical metric and a powerful feedback tool used to assess the effectiveness and quality of library resources and services within the education system (Riady et al., 2023; Kanwar and Sanjeeva, 2022). The satisfaction of the users will go a long way in making them feel that the library is an invaluable institutional resource, and it will contribute to loyalty, engagement, and an increase in the proportion of frequent use (Narasanna and Naik, R.

R., 2024; Gyau et al., 2021). Thus, user satisfaction as the main strategic priority, which can be attained through the ongoing evolution of services and updating of facilities, is necessary in order to address the new demands of the students and to establish the library as the inseparable support pillar of the academic community (Sumalinog et al., 2025).

Perceived Impact on Academic Productivity

Academic libraries are multifaceted in assisting and enriching the student learning and scholarly achievement through offering pertinent and precise information (Makhmaraimova et al., 2025). The impact of the library is spread both on the physical facility and the digital resources and services provided.

The physical environment is essential because it must be well managed, especially to provide areas of concentration and distraction-free conditions that would motivate students to excel in their studies. Moreover, the study proves that the use of libraries directly correlates with academic performance: the average GPA of students who used libraries in terms of entering the building, going to classes, or using online resources is higher (Scoulas and De Groote, 2022). It is supported by the results of the research that good learning environments, the attractiveness, organization, and customization have a positive impact on the possibility of students to study, sustain interest, and cooperate (Nja et al., 2023). The libraries should also make sure that they have a plethora of current and readily available materials such as books, journals, e-books, and websites because these materials and services (such as information and borrowing desks) are crucial to the ability of the students to perform research and complete school assignments (Arowolo et al., 2022; Fitria et al., 2024; Badr et al., 2024). The overall success of the students is proven to be correlated with the usage of the library resources (Mubofu and Malekani, 2021).

The modern scholarly environment uses technology as one of the main factors of library effectiveness. When the students are provided with the up-to-date technology and easily accessible interfaces, they are more involved and their learning process is improved with the possibilities of interactive and multimedia options (Pandita and Kiran, 2023). The high-paced utilization of library resources is not only achieved with the help of technological advances but also reinforced the management skills and eventually improved the performance of the users and their academic results (Khan et al., 2023). This is a crucial modernization since a large proportion of the respondents acknowledge that their library systems are useful and enhance their work, performance, productivity, and efficiency (Azeroual et al., 2019).

The libraries of the modern era need to be flexible, modernized, and open, with areas that are accommodative of different activities and modes of learning (Shaileshkumar & Bhatt, 2025). Through the adoption of new technology and student centricity, the libraries will also provide students with improved learning experience and stay abreast with the new fast changing technology (Mohammed and Alumona, 2025). Nevertheless, the barriers still persist, with some students struggling with the lack of knowledge, expensive internet access, or the help of the librarian (Mubofu and Malekani, 2021), which explains the importance of professional leadership and supportive services as well as the implementation of technology (Khan et al., 2023).

Age

Understanding different age groups is vital when researching library usage, as age significantly impacts people's perceptions and behavior (Bereens & Noorda, 2023). The authors noted that youth prefer libraries to be cozy and stress-free, and that these libraries should cater to both their social life and their studies. Similarly, Xu and Du (2019) found that age has a significant impact on users' satisfaction with digital libraries. Their analysis showed that graduate

students were more satisfied with the library's system and services than undergraduate students, meaning age influences how students view and expect library services.

School

Besides the students' ages, the schools they attend also have an impact on their experience with modern libraries. According to Rowe (2025), different schools have varying regulations and management structures for resources, which means that they function differently. These differences in access, budget, and priorities can determine the degree to which a library is modernized. Reis-Andersson (2024) pointed out that some schools can effortlessly implement digital systems and use modern tools, whereas others find it difficult. Similarly, Pritam and Shaikh (2024) clarified that while some schools have excellent technology, others face challenges with their facilities. It indicates that a school's capability to establish and sustain modern systems not only allows students to experience but also helps them understand the modernization of the library. As Sumartin (2021) affirmed, the extent to which a school cherishes its library can determine how students perceive and use it.

Year Level

A student's year level helps us understand their experiences and views because it changes as they progress through their studies. It is found that first-year students are less familiar with library resources and less prepared for college research compared to older students (LeMire et al., 2021). Furthermore, Dalal et al., (2022), also indicated that upper-year students are more skilled and better at using library resources. It shows that as students reach higher year levels, their experience, knowledge, and ability to use the library improve.

Frequency of Visiting Libraries

Students' year level is an essential factor in understanding their experiences and opinions, as it may differ depending on how long they have been studying. First-year students often hold different views and have less knowledge of university library resources, which is linked to their readiness for college research compared to higher-year students. On the other hand, Dalal et al. (2022) found that upper-year students typically have stronger skills and a greater ability to use library resources and systems. It shows that familiarity with the library varies among students by year level. Overall, this means that students' experiences in the library are influenced by how far they are in their studies, their skills, understanding, and how they manage their time.

1.3 Theoretical Framework

UX Theory

This study is anchored on the User Experience (UX) Theory by Norman and Nielsen (1998), which posits that user experience is shaped by all interactions a user has with a system, service, or environment. According to the theory, UX is influenced by several core dimensions: usability, usefulness, accessibility, emotional response, and the overall satisfaction derived from using a service. In this context, the academic library serves as the system through which students form their user experience based on the design, functionality, and quality of the services provided. The UX theory emphasizes that positive user experience emerges when a system effectively meets user needs, is easy to navigate, and provides both functional and emotional value.

Applying the UX Theory to the library setting, this study assumes that modernization efforts such as strengthening digital infrastructure, expanding access to online resources, automating library services, and improving physical learning spaces directly influence the quality of students' interactions with the library. These components act as the "design elements" of the library system, impacting how students perceive usability, usefulness, accessibility, and comfort. For example, efficient Wi-Fi, automated circulation systems, and updated study spaces may enhance students' sense of convenience and satisfaction, while abundant online resources contribute to perceptions of usefulness and academic support. Thus, library modernization serves as the primary input that shapes students' overall user experience.

Within this framework, students' library experience is understood as the outcome of how effectively modernization efforts improve UX dimensions. When modernization enhances usability, accessibility, and emotional satisfaction, students are more likely to report better ease of access to library services, greater satisfaction with facilities and digital systems, and stronger

perceptions of support for their academic activities. The UX Theory suggests that these positive experiences may also lead to increased engagement, higher frequency of library use, and overall improved academic support. Therefore, the theoretical framework positions library modernization as a key determinant of the students' library experience, with UX constructs serving as the mediating mechanism that explains how improvements in library systems translate into enhanced student outcomes.

Information System (IS) Success Model

DeLone and McLean Information Systems (IS) Success Model (1992; updated 2003), which explains how the quality of an information system influences user satisfaction, system use, and the overall benefits experienced by its users. The model identifies three major quality dimensions: System Quality, Information Quality, and Service Quality. In the context of library modernization, these dimensions correspond to institutional efforts to improve technologies, digital resources, automated processes, and learning environments within academic libraries.

In relation to the present study, System Quality is reflected in the modernization of digital infrastructure and the automation of library services, which affect students' ease of access and overall efficiency in using library tools. Information Quality is represented by the availability of online resources, which shapes students' perceived usefulness of modern library features that support their research and learning needs. Meanwhile, Service Quality aligns with the enhancement of physical learning spaces and updated service delivery, influencing students' satisfaction with the library environment and the support they receive during academic tasks.

The IS Success Model further posits that improvements across these quality dimensions lead to greater user satisfaction, more meaningful use of library systems, and higher perceived

net benefits. In this study, net benefits are reflected in students' perceived impact on academic productivity, showing how a well-modernized library contributes to better academic outcomes. Thus, the model supports the study's assumption that library modernization is significantly related to and can predict students' library experience, providing a strong theoretical foundation for analyzing how modernization efforts shape students' engagement with academic libraries.

Technology Acceptance Model (TAM)

Technology Acceptance Model (TAM) developed by Davis (1986), which explains how individuals come to accept and use new technologies. TAM proposes two core determinants of technology acceptance: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Perceived usefulness refers to the degree to which a person believes that using a system will improve their performance, while perceived ease of use refers to how effortless and user-friendly the system is. These constructs influence the user's attitude toward using the system, their behavioral intention to use it, and ultimately their actual usage.

In the context of library modernization, TAM aligns closely with the study's independent variables. Modernized features such as digital infrastructure, availability of online resources, and automation of library services directly influence students' perceptions of usefulness and ease of use. When library technologies are efficient, accessible, and reliable, students are more likely to perceive them as beneficial for research and academic tasks. Enhanced physical learning spaces also contribute indirectly by creating an environment that supports positive attitudes toward technology use, thereby strengthening students' intention to engage with modernized library systems.

TAM also provides a theoretical foundation for understanding the dependent variable, students' library experience. The model suggests that positive perceptions of ease of access, usefulness, and convenience lead to higher user satisfaction and increased system usage. In this study, these improved experiences translate into students' perceived impact on academic productivity, demonstrating how modernized library systems contribute to better academic outcomes. Thus, TAM supports the study's hypothesis that library modernization is significantly related to and may predict students' library experience, offering a strong theoretical basis for examining how technological advancements shape students' interactions with academic libraries.

1.4 Schematic Diagram

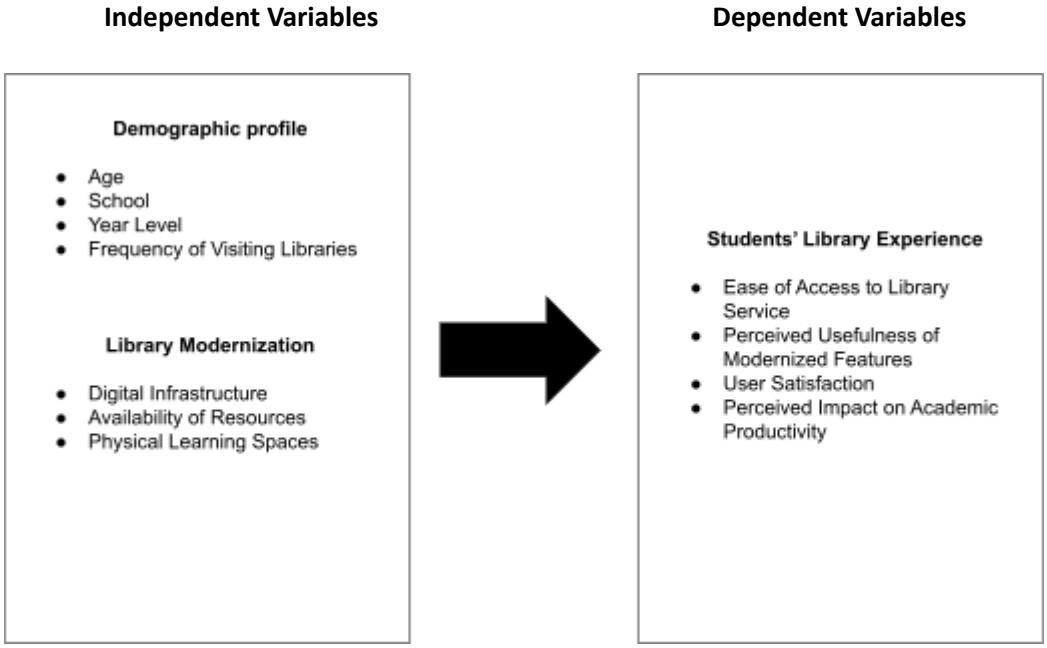


Figure 1. The Schematic Diagram shows the interplay between the independent and dependent variables

1.5 Statement of the Problem/ Objectives of the Study

The main goal of this study is to determine how modern libraries affect students' experiences in selected schools in Agusan del Norte. It also aims to identify the factors that make students satisfied with library services.

1. What is the Profile of the respondents in terms of:

   1.1 Age;

   1.2 School;

   1.3 Year Level;

   1.4 Frequency of Visiting Libraries

2. What is the level of library modernization in selected academic libraries in Agusan del Norte in terms of:

   2.1 Digital infrastructure,

   2.2 Availability of resources, and

   2.3 Physical learning spaces?

3. What is the level of students' library experience in terms of:

   3.1 Ease of access to library services,

   3.2 Perceived usefulness of modernized features,

   3.3 User satisfaction, and

   3.4 Perceived impact on academic productivity?

4. Is there a significant relationship between library modernization and students' library experience?

5. Is there a significant predictive relationship between library modernization and students' library experience among library users?

This study is important because it helps understand how modernizing the library affects students' experience and satisfaction. The goal of this research is to help improve library spaces and systems so they can better serve students' needs and support their academic success.

1.6 Hypotheses

Based on the research problems and objectives, the study follows this hypotheses:

H_o : There is a significant relationship between library modernization and students' library experience in selected academic libraries in Agusan del Norte

1.7 Scope and Limitations

The focus of this study is to understand how library modernization affects students' overall experience and satisfaction in academic libraries in Agusan del Norte. It also aims to understand how students feel about the library's facilities and services, including the space and environment, and how these factors influence their experiences.

This research targets students from different schools in Agusan del Norte who use their academic libraries. These students spend time in the library studying, doing schoolwork, finding learning materials, or relaxing. Their experiences are shaped by what the library offers, such as its services, resources, and environment. Overall, this study aims to better understand how students experience and benefit from modernized library systems and facilities.

The study focuses on college students from various universities in Agusan del Norte for the 2025–2026 school year. The primary data will be collected through survey questionnaires completed by consenting students. Topics unrelated to library modernization will not be part of this study.

The researchers will recruit participants from tertiary students across different programs at three universities in Agusan del Norte, with 50 students from each college.

To collect data, the researchers will use stratified sampling to reflect diverse student opinions. The Descriptive-Correlational Method will be used to analyze the relationship between students' experiences, satisfaction, and library modernization. By studying this relationship, the researchers aim to understand the collected data better and draw meaningful conclusions.

The study will be conducted during the academic year 2025–2026. By the end of 2026, the research will be completed, achieving its goals and objectives.

1.8 Definition of Terms

To help readers fully comprehend the terms used in this research, the following terms are defined conceptually and operationally.

Age - The specific number of years a respondent has lived, as recorded by the student in the demographic profile section of the paper-and-pencil survey.

Automation of Library Services - The implementation and utilization of computer software and systems to streamline and manage routine library operations (e.g., automated check-out, digital cataloging).

Availability of Online Resources - The extent to which the library provides and maintains a comprehensive collection of electronic materials, such as e-books, online databases, and digital journals, for student use.

Digital Infrastructure - The underlying technological framework, including networks, hardware, and core systems, that supports the delivery and access of the library's digital services and resources.

Ease of Access to Library Service - The calculated average score that measures the respondents' assessment of the simplicity and convenience in locating and utilizing the library's services, systems, and materials.

Enhanced Physical Learning Spaces - The improvements made to the library's physical environment, encompassing zones designed for individual quiet study, collaborative group work, and specialized learning activities.

Frequency of Visiting Libraries - The specific category of utilization (e.g., daily, weekly, monthly) selected by the student on the survey questionnaire indicating how often they physically visit the academic library.

Perceived Impact on Academic Productivity - The calculated average score that measures the respondents' belief that the library's services and resources, particularly the modernized features, contribute to increased efficiency, output, and quality of their academic work.

Perceived Usefulness of Modernized Features - The calculated average score reflecting the respondents' conviction that the new or upgraded features (e.g., technology, services, and spaces) provided by the library are effective, helpful, and contribute positively to their academic studies.

School - The name of the specific academic institution (Saint Michael College of Caraga, Father Saturnino Urios University, or Caraga State University) where the respondent is a currently enrolled student.

User Satisfaction - The calculated overall average score indicating the degree to which respondents feel a sense of contentment, happiness, and fulfillment with their entire library experience, including staff, resources, and facilities.

Year Level - The specific academic standing (First Year, Second Year, Third Year, or Fourth Year) selected by the respondent in the demographic section of the survey.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter represents the research design, the research materials and methods, Environment, population and sample, and the ethical standards that will be used in this research study.

2.1 Research Design

This study will use a descriptive-correlational quantitative research design. It aims to determine how library modernization relates to students' library experiences in selected schools in Agusan del Norte. It will also look at how modern facilities, digital services, and access to technology affect student satisfaction and library use. This design helps the researchers describe and measure the relationship between modernization and students' experiences.

2.2 Research Materials and Methods

The main research instrument will be a structured survey questionnaire, adapted and modified from previously validated studies to align with the objectives of this research. The questionnaire consisted of three main parts:

Part I. Demographic Profile - age, school, year level, and frequency of library visits.

Part II. Library Modernization - digital infrastructure, availability of online resources, automation of library services, and enhancement of physical learning spaces.

Part III. Students' Library Experience - measured through indicators such as ease of access to library services, perceived usefulness of modernized features, user satisfaction, perceived impact on academic productivity.

A five-point Likert Scale will be used to assess students' responses regarding their satisfaction with library services. This scale will serve as the primary psychometric tool to quantify attitudes and opinions, allowing the researchers to measure the degree of satisfaction.

Scale	Verbal Description
5	**Very Satisfied**
4	**Satisfied**
3	**Neutral**
2	**Dissatisfied**
1	**Very Dissatisfied**

Table 2: Shows the scale description of the survey questionnaire of the study

In addition to the main questionnaire, a baseline survey—the Library Information Form—will be administered to the participating head librarians of academic libraries within Agusan del Norte.

The library information form is divided into the following sections: (1) general information; (2) library facilities and technologies; (3) library policies and programs; (4) accessibility and operations; (5) additional information. This form aims to gather

essential information regarding the modernization of libraries and to ensure that each institution meets the study's selection criteria for modernized academic libraries.

1. **Location:** Library must be located within Agusan del Norte.

2. **Type of institution:** A library must be part of a higher education institution (college or university).

3. **Modernized Features:** Library must have adopted modernization initiatives such as:

    - Online Public Access Catalog (OPAC)
    - Access to E-Resources and databases
    - Computerized circulation or RFID systems
    - Technology-integrated facilities (computers, multimedia rooms, Wi-fi access)
    - ICT-based library services or digital reference systems.

4. **Accessibility and Cooperation:** The institution must grant permission and allow researchers to conduct surveys or observations within the library premises.

5. **Active Student Users:** Library must have regular student users who utilize the library premises

6. **Willingness to Participate:** Library administrators and staff must be willing to cooperate with the research team.

7. **Operational Readiness:** The library must be operational during the period of data collection.

Aside from the primary tool, a Library Information Form will be given to the head librarians of each school. This form will collect information about the library's facilities, technology, rules, accessibility, and modernization programs. It will also make sure that only libraries with an OPAC, e-resources, ICT services, and WiFi are included in the study.

2.3 Research Environment, Population, and Sample

The research will be conducted at three learning institutions in Agusan del Norte. Saint Michael College of Caraga (SMCC), Father Saturnino Urios University (FSUU), and Caraga State University (CSU). These schools will be selected because they have modernized academic libraries that include digital materials and technology-based services.

The research sample consisted of college students who actively used the library throughout the 2025-2026 academic year. The method used was stratified random sampling to select them, and the three institutions ensured proportional representation. A total of 108 students were involved, equally in the schools:

RESPONDENTS (Library Users)	Population	Sample	
	F	f	&
CSU	50	36	33.3
FSUU	50	36	33.3
SMCC	50	36	33.3
Total	**150**	**108**	**100.0**

The sample will be derived proportionally to ensure equal representation of each institution's population. Only students with prior experience using both traditional and modernized library systems were included to provide relevant responses.

2.4 Research Data and Analysis Plan

Before collecting data, the researchers will first get formal letters and permits from the heads of the academic institutions involved. They will also coordinate with the PLAI–Caraga Region Librarians Council to confirm the list of approved libraries in Agusan del Norte. After that, consent will be obtained, and the survey will be given to the chosen participants.

The researchers also communicated with the Secretary of the PLAI–Caraga Region Librarians Council to request a list of accredited academic libraries for the study. Based on the information received, they created a baseline survey and prepared formal request letters, signed by the Research Adviser and the Dean, to allow the research to proceed.

To analyze data, descriptive and correlational statistics methods will be used. Data on the demographic Profile of respondents and the level of satisfaction were described using descriptive statistics (frequencies, percentages, means, and standard deviations).

The correlation analysis used the mean to assess the relationship between the modernization of library facilities and students' total library experience.

Coding of data will ensure consistency and adoption of statistical software, enabling easy computation and interpretation.

2.5 Research Ethical Standards

This study is guided by ethical principles that ensure all methods are practiced with integrity, accountability, and respect to participants. These principles serve as a foundation to ensure public trust in the research study's findings. The core ethical principles embraced in this chapter are integrity, honesty, informed consent, confidentiality, transparency, avoidance of plagiarism, and responsible use of technologies.

A. Intellectual Property Rights:

This study will adhere to all principles of Intellectual Property Rights. All materials created for the research, such as the surveys and data collection instruments, will be protected under Philippine copyright law. Any information derived from other sources, such as books or studies, will be properly cited to respect the original authors' work and protect the researchers' own contributions. Since the study does not involve creating a brand, logo, or symbol, no trademark registration will be needed.

B. Informed Consent:

Participants, who will be college students from three schools in Agusan del Norte, will be clearly told about the study's purpose, procedures, benefits, and risks before they receive the survey. Written consent in simple language will be obtained, and students will be informed that they can withdraw at any time without consequences. No animal testing will be involved, so no extra ethical approval will be needed.

C. Data Privacy and Confidentiality:

All personal information is collected only with consent and will be handled in accordance with the Data Privacy Act of 2012 (RA 10173). The data is used only for this study and will not be shared carelessly. Participants can choose to give their name, but anonymity is ensured. Only researchers can access the data, which is safely stored both physically and digitally. Anonymized data may be shared with third parties with participants' consent, in accordance with ethical and legal rules.

D. Voluntary Participation:

Participants are randomly selected and may refuse to participate in the survey. Participation is entirely voluntary, and they can leave at any time without penalties. No animals are involved, so no special ethical rules apply.

E. Minimization of Harm:

The study is designed to prevent physical, emotional, or psychological harm. Participants are told they can stop if they feel uncomfortable. Support services are available if needed. No animals are involved, so no extra precautions are required.

F. Beneficence:

The study benefits the academic community and society by providing insights into student satisfaction with library modernization. Findings can guide future research and help schools improve their services. Participants can request a copy of the study results to acknowledge their contribution.

G. Justice and Fair Participant Selection:

Participants are randomly selected fairly from the schools. Any student using the libraries can participate without discrimination. Since no animals are involved, no ethical rules about species are required.

H. Data Integrity and Accuracy:

Standard methods were used to collect and analyze data. Results were carefully checked to reduce errors and biases. Study limitations, such as sample size and potential bias, are acknowledged.

I. Transparency and Honesty in Reporting:

The findings will be reported honestly without altering data. All sources and materials are properly cited to respect intellectual property rights.

J. Use of Patented or Copyrighted Materials:

No patented technologies will be used. Copyrighted materials, like books or studies, will correctly cited to support the research.

K. Ethical Considerations for Human and Animal Trials:

Participants will be informed of the risks and benefits, as well as their right to withdraw. Since no animals will be involved, no additional ethical approval will be required.

L. Responsible Use of AI and Related Technologies:

AI tools and digital technologies were used responsibly. Grammarly helped with grammar, and ChatGPT helped organize ideas. Researchers ensured that the tools did not provide false or biased information.

M. Ethical Clearance and Institutional Approval:

Ethical approval will be obtained from the Institutional Review Board of Saint Michael College of Caraga, ensuring compliance with all rules regarding participant protection, data privacy, and intellectual property rights.

REFERENCES

Abouelela, A. (2022). Towards a Better Interior Design for the Academic L ibrary at College of Education—King Faisal University. *Designs, 6(3), 47.* https://doi.org/10.3390/designs6030047

Abrigo, C. M., Lyn, D., & Labangon, G. (2021). The Library as a Digital Scholarship Hub: Opportunities for Leveraging Learning Support. *Qualitative and Quantitative Methods in Libraries (QQML)*, *10*, 585–608.

Acta Informatica Malaysia (AIM), (2023). *LIBRARY AND INFORMATION SERVICES IN THE DIGITAL AGE: OPPORTUNITIES AND CHALLENGES.* https://actainformaticamalaysia.com/archives/AIM/2aim2023/2aim2023-113-121.pdf

Agina-Obu, R., & Sawyerr-George, O. E. (2023). Evaluation of Users' Satisfaction of Information Resources in University Libraries in Nigeria: A Case Study. *Indian Journal of Information Sources and Services*, *13*(1), 1–5. https://doi.org/10.51983/IJISS-2023.13.1.3378

Ajani, O., & Buraimo, A. (2022). A proposed computer vision-based system for managing the self-borrowing and returning process in libraries. *International Journal of Library and Information Sciences.* https://doi.org/10.21608/ijlis.2023.239371.1216

Alkali, J. A., & Muhammed, B. U. (2024). User satisfaction with academic library services; An empirical review. *Global Scientific JOURNALS*, *12*(5), 1011–1018. Retrieved from http://www.globalscientificjournal.com

Amosi, M. K. (2020). *CHALLENGES OF COMPUTER NETWORK INFRASTRUCTURE IN ACCESSING ELECTRONIC INFORMATION RESOURCES IN ACADEMIC LIBRARIES.* http://repository.out.ac.tz/2972/1/MUSA%20KASELLA%20AMOSI%20tyr%20haki.pdf

Armilla, R. R., & Abangan, M. L. M. (2025). The Difficulties and Strategies in Digitizing Library Resources. *SSRN Electronic Journal*. https://doi.org/10.2139/SSRN.5134260

Arowolo, R. O., Ajiboye, B. A., & Adio, S. K. (2022). Utilisation of Library information resources by Undergraduates of Public University in Ogun State. *Library Philosophy and Practice (e-Journal)*. https://digitalcommons.unl.edu/libphilprac/7451

Ashikuzzaman, M. (2025, August 6). The future of Public Libraries: Adapting to the needs of modern communities. *Library & Information Science Education Network.* https://www.lisedunetwork.com/the-future-of-public-libraries-adapting-to-the-needs-of-modern-communities/

Azeez, A., & Olanrewaju, A. O. (2019). Use of Technology Acceptance Model (TAM) to evaluate library electronic information resources use by undergraduate students of Lead City University, Ibadan, Nigeria. *Library Philosophy and Practice.* https://www.researchgate.net/publication/333103888_Use_of_Technology_Acceptance_Model_TAM_to_Evaluate_Library_Electronic_Information_Resources_use_by_Undergraduate_Students_of_Lead_City_University_Ibadan_Nigeria

Azeroual, O., Schöpfel, J., & Saake, G. (2019). Implementation and User Acceptance of Research Information Systems An Empirical Survey of German Universities and Research Organisations. https://dspacecris.eurocris.org/

Badr, A. M. M., Al-Abdi, B. S., Rfeqallah, M., Kasim, R., & Ali, F. A. M. (2024). Information quality and students' academic performance: the mediating roles of perceived usefulness, entertainment and social media usage. *Smart Learning Environments*, *11*(1), 1–20. https://slejournal.springeropen.com/articles/10.1186/s40561-024-00329-2

Baffour Gyau et al., (2021), Evaluation of User Satisfaction with Academic Libraries Services Based on Students' Perspectives. Open Access Library Journal, 8: e7783. https://doi.org/10.4236/oalib.1107783

Bagewadi, *P., Mathapati, D., Gangur, B., & Patil, S. S. (2025).* Automating libraries: The role of library management systems in modern library operations. *International Journal of Research in Library Science, 11(2), 105–114. https://doi.org/10.26761/ijrls.11.2.2025.1877*

Banleman, K. B., Banleman, B. K. D., & Banleman, L. D. (2023). ACADEMIC LIBRARIES AND DISTANCE EDUCATION IN GHANA: EVIDENCE FROM THE SD DOMBO UNIVERSITY OF BUSINESS AND INTEGRATED DEVELOPMENT STUDIES. *International Journal of Applied Research in Social Sciences*, *5*(10), 422–439. https://doi.org/10.51594/IJARSS.V5I10.632

Barfi, K. A., Parbie, S. K., Filson, C. K., Teye, M. V., Kodua-Ntim, K., & Ayensu, E. (2023). Assessing the quality of services at an academic library. *Heliyon*, *9*(12), e22449. https://doi.org/10.1016/j.heliyon.2023.e22449

Basiag, M. (2025). Student perspectives on the ease of use and acceptance of electronic resources. *Journal of Interdisciplinary Perspectives*, *3*(7). https://doi.org/10.69569/jip.2025.290

Berni, A., & Borgianni, Y. (2021). From the definition of user experience to a framework to classify its applications in design. *In Proceedings of the Design Society: Volume 1 – ICED 21 (pp. 1627–1636). Cambridge University Press.* https://doi.org/10.1017/pds.2021.424

Burhansab, P. A., Sadik Batcha, M., Ahmad, M., & Batcha, S. (2020). Impact of Web 2.0 Technologies on Academic Libraries: A Survey on Affiliated Colleges of Solapur University. *Library Philosophy and Practice (e-Journal)*. https://digitalcommons.unl.edu/libphilprac/4587

Cabonero, D. A., Austria, R. M., Mark, J., Bayang, L., & Bumanghat, R. P. (2020). Knowledge, Satisfaction and Challenges on Online Public Access Catalog in an Academic Library in Northern Philippines. *Library Philosophy and Practice (e-Journal)*. https://digitalcommons.unl.edu/libphilprac/3852

Carvalho Rodrigues, M., Mandrekar, B., Carvalho Rodrigues Librarian, M., & Mandrekar Librarian, B. (2020). IMPACT OF ACADEMIC LIBRARY SERVICES ON STUDENTS SUCCESS AND PERFORMANCE. *Library Philosophy and Practice (e-Journal)*. https://digitalcommons.unl.edu/libphilprac/4246

Cohen, A. (2025). Chapter 5 - Modernizing the library program. https://www.sciencedirect.com/science/chapter/monograph/abs/pii/B9780443291760000051

Corpuz, D. A. (2020). LIBRARY RESOURCES AND FUNCTIONAL EFFECTIVENESS OF AN ACADEMIC LIBRARY: MEETING THE CHALLENGES OF THE DIGITAL AGE. *Humanities & Social Sciences Reviews*, *8*(4), 238–245. https://doi.org/10.18510/hssr.2020.8425

Cui, X., & Ahn, C. (2024). The mediating role of space satisfaction in enhancing learning outcomes: A study of university library learning spaces and informal learning needs. *International Journal of Sustainable Building Technology and Urban Development*, 545–562. https://doi.org/10.22712/susb.20240038

Cui, X., Liao, J., & Ahn, A. C. (2025). Scientometric review of informal learning spaces in university libraries: A bibliometric approach to design and trends. *Sustainability, 17(6), 2480.* https://doi.org/10.3390/su17062480

DeLone, W. H., & McLean, E. R. (1992). Information Systems Success: The Quest for the Dependent Variable. *Https://Doi.Org/10.1287/Isre.3.1.60*, *3*(1), 60–95. https://doi.org/10.1287/ISRE.3.1.60

Diller, K. R., & Wallin, S. B. (2023). *Place Attachment, Libraries, and Student Preferences* (Vol. 23, Issue 4).

Dianing, P. J. A., & Acedera, A. P. (2025). Library design and academic performance: Implications on students' engagement. *International Journal of Current Science Research and Review, 8*(5), 2251–2262. https://doi.org/10.47191/ijcsrr/V8-i5-33

Donovan, J. (2020). Keep the books on the shelves: Library space as intrinsic facilitator of the reading experience. *The Journal of Academic Librarianship*, 46 (2), 102104. https://doi.org/10.1016/j.acalib.2019.102104

Dorado, D. A. D. (2019). *View of A STUDY ON THE PERCEPTION OF FILIPINO LIBRARIANS ON INNOVATION*. https://phjlis.org/index.php/phjlis/article/view/24/19

Fagyan, C. K., Macalingay, M. B., Abellada, A. P., Munar, B. B., Depnag, M. C., Bheal, A., & Kitani, M. (2023). *The Impact of Library Use Frequency on Student Satisfaction: An Evaluation of Resources, Services, and Facilities*. https://www.irejournals.com/formatedpaper/1704885.pdf

Fernandez, G. M. J., Nuera, R. E., Tianio, J. S., & de Castro, J. M. (2025). *INFORMATION TECHNOLOGY, INTERPERSONAL, AND LEADERSHIP COMPETENCIES OF 21ST CENTURY FILIPINO LIBRARIANS AND THEIR IMPACT ON ACADEMIC LIBRARY SERVICES*. https://doi.org/10.5281/ZENODO.15344874

Fitria, Yahya, M., Ali, M. I., Purnamawati, & Mappalotteng, A. M. (2024). The Impact of System Quality and User Satisfaction: The Mediating Role of Ease of Use and Usefulness in E-Learning Systems. *International Journal of Environment, Engineering and Education*, *6*(2), 119–131. https://doi.org/10.55151/IJEEDU.V6I2.134

Fufuri, ibrahim, Phd, (2024) IBRAHIM M. M. FURFURI, (PhD), (2024). *Digital Age and Academic Libraries in Nigeria: The Emerging Roles of Academic Librarians*. https://kijcl.khairun.edu.ng/index.php/kijcl/article/view/41/36

Galagala, M. R. G. (2024). *University students' satisfaction with library services and resources using the Kano model*. https://doi.org/https://doi.org/10.21833/ijaas.2024.05.009

Galang, G. M. B., & Villa, E. B. (2024). Philippine Social Security System: An Evaluation Of Strategic Approach To Digital Services. *International Journal of Multidisciplinary: Applied Business and Education Research*, *5*(8), 3272–3300. https://doi.org/10.11594/IJMABR.05.08.26

Goodluck Ifijeh & Felicia Yusuf, (2020). *Covid – 19 pandemic and the future of Nigeria's university system: The quest for libraries' relevance*. https://www.sciencedirect.com/science/article/pii/S0099133320301178

Gore, V. (2023). The impact of library automation on user satisfaction and service efficiency. *International Journal of Engineering Research and Management Technology, 10*(4), 309–313. https://ijermt.org/publication/68/109.%20july%202023%20%20ijermt.pdf

Guchacha, J. (2019). Integration of cloud computing and service delivery in academic libraries: A case of South Eastern Kenya University (Doctoral dissertation, University of Nairobi). *University of Nairobi Repository.* https://tinyurl.com/5ejw3xtw

Guilles, D. M. (2019). *RECONSTRUCTING THE PAST: HISTORY OF MUNICIPAL, CITY, AND PROVINCIAL LIBRARIES IN THE PROVINCE OF RIZAL FROM 1951 TO 2018*. https://tinyurl.com/nadup8wb

Gyau et al., (2021) International Students Library Usage Frequency Patterns in Academic Libraries: A User Survey at Jiangsu University Library. *Open Access Library Journal, 8, Article No. e7610.* https://doi.org/10.4236/oalib.1107610

Hussain, M., & Ameen, L. (2023). Exploring the automation landscape in university libraries of Khyber Pakhtunkhwa, Pakistan: Status, problems and implications. *The Journal of Academic Librarianship, 49(6), 102787*. https://doi.org/10.1016/j.acalib.2023.102787

Innovation Acceptance Lab. (2020, May 23). Technology Acceptance Model (TAM) - Technology Acceptance Lab. *Technology Acceptance Lab.* https://acceptancelab.com/technology-acceptance-model-tam

Iroroeavwo, A. E., Samuel, E. ., & Chioma, A. A. . (2024). Accessibility and use of electronic resources for digital reference services among university libraries in southern Nigeria. *Regional Journal of Information and Knowledge Management*, *8*(2), 1–16. Retrieved from https://www.ajol.info/index.php/rjikm/article/view/263049

Ivanova, V. (2024). Automation of Library Services—Turning Point in Development of Academic Libraries. In Proceedings of the International Conference on Electronics, Engineering Physics and Earth Science (EEPES 2024) (Vol. 70, No. 1, p. 38). *MDPI. https://doi.org/10.3390/engproc2024070038*

Jimenez, S. A., Berbegal-Mirabent, J., & de la Torre, R. (2024). How do university libraries contribute to the research process? *The Journal of Academic Librarianship*, *50*(5), 102930. https://doi.org/10.1016/J.ACALIB.2024.102930

Jumino, J., & Kusumarini, N. I. (2024). *Analysis of the Interior Design of the Library, Faculty of Humanities, Diponegoro University*. 59–67. https://doi.org/10.2991/978-2-38476-313-9_10

Kabatangare T. G. (2019). *Transforming Library and Higher Education Services in Support*

of Industrialization and Modernization On Staying in the Game: Redefining Academic Library Service to Meet 21st Century User Needs https://backend.kab.ac.ug/server/api/core/bitstreams/f0bff436-bb5f-4dbe-ac6d-0759386ad8d6/content

Kadir, M. R. A., Samsudin, A. Z. H., & Seman, M. R. (2023). Analysis of User Satisfaction of Academic Digital Library: An Empirical Model with Mediating Effects. *International Journal of Academic Research in Business and Social Sciences*, *13*(12). https://doi.org/10.6007/IJARBSS/V13-I12/20030

Kanwar A and Sanjeeva M (2022). Student satisfaction survey: A key for quality improvement in the higher education institution. *Journal of Innovation and Entrepreneurship, 11(1): 27.* https://doi.org/10.1186/s13731-022-00196-6

Kapadi, R. V., & Patil-Dalve, D. (2023). *Collection development policies in the digital age*. *Aegaeum, 11*(8), 36–43. Retrieved from https://www.researchgate.net/publication/373216751_Collection_Development_Policies_in_Digital_Age

Kaushamalika, M. K., & Weerakoon, W. R. W. M. A. U. (2020). Students' satisfaction with library services and facilities at three regional center libraries of the Open University of Sri Lanka. *Journal of the University Librarians Association of Sri Lanka*, *23*(1), 37–56. https://doi.org/10.4038/jula.v23i1.7965

Khan AU, Rafi M, Zhang Z, Khan A (2023), "Determining the impact of technological modernization and management capabilities on user satisfaction and trust in library services". *Global Knowledge, Memory and Communication*, Vol. 72 No. 6-7 pp. 593–611, doi: https://doi.org/10.1108/GKMC-06-2021-0095

Kumar, R. (2023). *Students' perspectives on the application of Internet of Things for redesigning library services at Kurukshetra University*. *Open Information Science, 7*(1), Article 20220154. https://doi.org/10.1515/opis-2022-0154

Kuyela, K., Mvula, D., & Muntanga, E. (2025). An Assessment of the Effect of the Fourth Industrial Revolution on Library Services: A Case of Academic Libraries in Lusaka Province. *International Journal of Research and Innovation in Social Science*, *IX*(VII), 3957–3970. https://doi.org/10.47772/IJRISS.2025.907000319

Lascano, E. A., & Lascano, E. A. (2021). Students' Knowledge on Library and Information Science (LIS): An Option for Choosing a Career. *Library Philosophy and Practice (e-Journal)*. https://digitalcommons.unl.edu/libphilprac/5875

Lasig, C. A. (2024). Students' Awareness and Use of the Online Public Access Catalog (OPAC) at the Central Luzon State University Library in the Philippines. *International Journal of Information Science and Management (IJISM)*, *22*(2), 77–88. https://doi.org/10.22034/IJISM.2024.1977926.0

Leonido, M., Melchor, A. F. E., San Pedro, J. R., & Vasquez, C. (2024). *Assessment of Library Resources, Services, and Facilities in the Digital Age Basis for a Development Plan*. https://tinyurl.com/4fjjszb

Maraon, M. I., & Agcito, M. R. (2024). *EXPLORING LIBRARIANS' KNOWLEDGE, TECHNOLOGY ACCEPTANCE, AND UTILIZATION OF EMERGING TECHNOLOGIES*. https://doi.org/10.5281/ZENODO.11189661

Mahanta, Prafulla Kumar (2019). *ICT Infrastructure and Services in the College Libraries of Assam: A Study.* *https://tinyurl.com/afwr548j*

Makhmaraimova, S., Mustafaeva, N., Buriyeva, L., Ashirbaeva, D., Kilicheva, K., Rustamova, M., Sapaev, I. B., & Turakulova, A. (2025). Information Services in School Libraries: A Study of their Contribution to Student Academic Development. *Indian Journal of Information Sources and Services*, *15*(2), 413–419. https://doi.org/10.51983/IJISS-2025.IJISS.15.2.50

Mehta, P., & Cox, A. (2021). At Home in the Academic Library? A Study of Student Feelings of "Homeness." *New Review of Academic Librarianship*, *27*(1), 4–37. https://doi.org/10.1080/13614533.2018.1547774

Mir, S. A. (2024). *Library automation: Enhancing efficiency and user experience*. *International Journal of Research Publication and Reviews, 5*(3), 1354–1357. https://ijrpr.com/uploads/V5ISSUE3/IJRPR23461.pdf

Mohammed, L. G., Alumona, A. I., & Uzomba, C. E. (2025). MODERNIZING LIBRARY SERVICES IN THE DIGITAL ERA: NEW TRENDS AND USER-CENTRIC CHALLENGES. *Journal of Applied Information Science & Technology*, *18*(1), 188–200. https://doi.org/10.70118/jaist.202501801.18

Moonasar, Anushie (2024). *Assessing the alignment of a University of Technology Library infrastructure with the Fourth Industrial Revolution: implications for librarian roles.* *https://www.emerald.com/lm/article-pdf/46/1-2/94/10064300/lm-06-2024-0069.pdf*

Mubofu, C., & Malekani, A. (2021). Accessibility of Library Resources and Support Services by Distance Learners. *Journal of Library & Information Services in Distance Learning*, *15*(4), 267–279. https://doi.org/10.1080/1533290X.2021.2021345

Musa, M., Darman, & Yunus, S. (2023, December). The effect of ease of access to services and transparency of information on population service satisfaction. *West Science Interdisciplinary Studies,* *1*(12), 1233–1240. https://pdfs.semanticscholar.org/5f49/0df783839160aa3f0bdda57b8f62a130eded.pdf

Nakhumicha, T. A. (2024). Access and utilization of information in digital resource platforms for teaching and research by academic staff in selected public universities, Kenya (Master's thesis). *University repository.* http://localhost:8080/xmlui/handle/123456789/5722

Narasanna, & Naik, R. R. (2024). Users satisfaction of library resources and services at university of horticulture science, Bagalkote, Karnataka. *World Journal of Advanced Research and Reviews*, *24*(02), 2717–2727. https://doi.org/10.30574/wjarr.2024.24.2.3194

Nja, C. O., Anari, M. I., Erim, C. M., Idiege, K. J., Ilhami, A., Ukah, J. U., Eneyo, O. E., Uwe, U. E., & Cornelius-Ukpepi, B. U. (2023). Learning space, students' collaboration, educational outcomes, and interest: Exploring the physical, social and psychological mediators. *Heliyon*, *9*(4), e15456. https://doi.org/10.1016/j.heliyon.2023.e15456

Odunola, O. A., & Tella, A. (2020). *Perceived usefulness as correlate of undergraduate students'*

patronage of online public access catalog in South-West Nigeria. *Cataloging & Classification Quarterly, 58*(5), 520–532. https://doi.org/10.1080/01639374.2020.1765443

Osisanwo, T. A., & Adeola, O. (2024). Perceived ease of use of innovative library practices for service delivery by librarians in public institutions in Ogun State. *The Catalyst Journal of Library and Information Literacy, 3*(1), 174–186. https://journals.journalsplace.org/index.php/CJLIL/article/viewFile/542/468

Pandita A and Kiran R (2023). The technology interface and student engagement are significant stimuli in sustainable student satisfaction. Sustainability, 15(10): 7923. https://doi.org/10.3390/su15107923

Pradhan, S., & Shrestha, K. (2024). Digital literacy in academic libraries: A systematic review. *Library Philosophy and Practice (e-journal)*, *28*(1), Article 11550. https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=11550&context=libphilprac

Rajwanti. (2025). *Library automation in academic libraries: An overview. All Research*, 11(4B). https://doi.org/10.22271/allresearch.2025.v11.i4b.12464

Riady Y, Sofwan M, Mailizar M, Alqahtani TM, Yaqin LN, and Habibi A (2023). How can we assess the success of information technologies in digital libraries? Empirical evidence from Indonesia. *International Journal of Information Management Data Insights, 3(2): 100192.* https://doi.org/10.1016/j.jjimei.2023.100192

Rodenbarger, L., & Hortsch, M. (2020). How students choose e-learning resources: The importance of ease, familiarity, and convenience. *FASEB BioAdvances, 2*(2), 90–101. https://doi.org/10.1096/fba.2019-00094

Sayekti, R., Aditya, M., Nurhayani, Simahate, T., Yusniah, & Devianty, R. (2022). An assessment of information control: Understanding library service quality from users' perspectives. *Evidence Based Library and Information Practice, 17*(2), 88–108. https://doi.org/10.18438/eblip29916

Scoulas, J. M., & de Groote, S. L. (2022). *Impact of Undergraduate Students' Library Use on Their Learning beyond GPA: Mixed-Methods Approach*. https://crl.acrl.org/index.php/crl/article/view/24743/33322

Shaileshkumar, P. P., Dhavalben, D., & Bhatt, D. (2025). Innovation in University Libraries: Creating Spaces that Foster Academic Growth. In *© 2025 IJRTI* | (Vol. 10, Issue 2). www.ijrti.org

Shee, S. (2025). Library automation in the digital age: Trends and challenges. *International Research Journal of Humanities & Interdisciplinary Studies, 6*(4), 75-84. https://doi-ds.org/doilink/04.2025-36677948/IRJHIS2504009

Somaratna, S. D. (2022). Planning and designing learning spaces in academic libraries: A case study at the Information & Learning Centre (ILC) of the Faculty of Science, University of Colombo. *Journal of the University Librarians Association of Sri Lanka, 25*(2), 1–33. https://doi.org/10.4038/jula.v25i2.8055

Sousa, N. M. T. (2025). *Academic libraries as hubs of artificial intelligence competency.* *https://link.springer.com/article/10.1007/s44163-025-00490-8*

Suharto, S. S. & Abdul Kadir, M. R. (2021). User Satisfactions on Digital Library: A Correlational Study. *Journal of Information and Knowledge Management (JIKM), 11(2).* https://ijikm.uitm.edu.my/pdf/1129.pdf

Sumalinog, L. C., Redondo, R. P., & Tagarda, E. P. (2025). Use and Quality of Library Services: Implications on the Satisfaction of Students' Information Needs. *British Journal of Multidisciplinary and Advanced Studies, 6(1), 1–12. https://doi.org/10.37745/BJMAS.2022.04228*

Surairi, M., Adam, M., & Mukhlis. (2025). *The effect of perceived usefulness and perceived ease of use on technology usage and its impact on technology adoption of new PLN Mobile App: Study on PLN UP3 Banda Aceh Office customer. International Journal of Scientific and Management Research, 8*(2), 73–83. https://doi.org/10.37502/IJSMR.2025.8207

Suresh, P. (2025.). Library automation and library management essential in the digital era. *Department of Library & Information Science, S. K. University, Ananthapuramu.* https://ijrcs.org/wp-content/uploads/IJRCS202509006-min.pdf

Goodluck Ifijeh & Felicia Yusuf, (2020). *Covid – 19 pandemic and the future of Nigeria's university system: The quest for libraries' relevance*. https://www.sciencedirect.com/science/article/pii/S0099133320301178

Ternenge, T. S., & Kashimana, F. (2019). Availability, accessibility, and use of electronic information resources for research by students in Francis Sulemanu Idachaba Library, University of Agriculture, Makurdi. *Library Philosophy and Practice, Article 2352*. University of Nebraska–Lincoln. https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=5768&context=libphilprac

Treceñe, J. K. D. (2021). The Digital Transformation Strategies of the Philippines from 1992 to 2022: A Review. *Engineering & Technology Review*, *2*(1), 8–13. https://doi.org/10.47285/etr.v2i1.66

Tuble, R. M., Johanna, M., & Bayoneta, A. R. (2019). The Performance of Public Libraries in Negros Occidental, Philippines. *In Philippine Social Science Journal* (Vol. 2). https://orcid.org/0000-0002-8204-0107

Umoh, M. S., & Agwunobi, J. N. (2020). Users' satisfaction with facilities of university libraries in South-South, Nigeria. *Journal of Library Services and Technologies*, *2*(1), 69–78. https://doi.org/10.47524/jlst.v2i1.7

Upadhayay, D. (2020). *Digital library: An Innovation in Infrastructure in Learning and Development amongst Students.* http://bit.ly/49QbMri

Vaish, Mamta (2024). ROLE OF ACADEMIC LIBRARIES IN THE DIGITAL ENVIRONMENT: A STUDY https://www.klibjlis.com/11.1.1.pdf

Waray, P. (2025). Library automation in the present scenario of colleges: Processes, benefits, and challenges. *SVKT Arts, Science and Commerce College.https://ijmrtjournal.com/wp-content/uploads/2025/05/Library-Automation-1.pdf*

Winata, A. P., Fadelina, R., & Basuki, S. (2021). New normal and library services in Indonesia: a case study of university libraries. *Digital Library Perspectives*, *37*(1), 77–84. https://doi.org/10.1108/DLP-07-2020-0059

Xie, I., & Babu, R. (2020). Enhancing usability of digital libraries: Designing help features to support blind and visually impaired users. *Information Processing & Management, 57*(3), 102245. https://doi.org/10.1016/j.ipm.2019.102245

Yarima, F. (2021). Motivational Factors for Users Satisfaction with Library and Information Services in Federal University Library, Lokoja. Logo State. *www.academia.edu*. https://www.academia.edu/download/93032872/483651866.pdf

Zhu, X.A., Whitaker, E. E., Cho, M., & Zhang, M. (2025). Student Engagement in Academic Libraries: A Conceptual Framework. *College & Research Libraries,* 86(2), 275. https://doi.org/10.5860/crl.86.2.275

CURRICULUM VITAE

NAME: ANGEL MICA J. GONZALES
ADDRESS: BRGY 4, NASIPIT AGUSAN DEL NORTE
EMAIL: angelmica_gonzales@smccnasipit.edu.ph



Personal Information

Sex	:	Female
Date of birth	:	January 4, 2005
Height	:	150 cm
Weight	:	68
Civil Status	:	Single

Educational Attainment

Elementary : Nasipit Central Elementary School
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Future Librarians and Information Scientists' Society (S.Y 2025-2026)	Public Information Officer
Future Librarians and Information Scientists' Society (S.Y 2024-2025)	Business Manager

CURRICULUM VITAE



NAME: ASHLLY J. LLAVAN
ADDRESS: D-NARRA, BRGY 1, NASIPIT, AGUSAN DEL NORTE

EMAIL: ashlly_llavan@smccnasipit.edu.ph

Personal Information

Sex	:	Female
Date of birth	:	February 15, 2005
Height	:	152 cm
Weight	:	42
Civil Status	:	Single

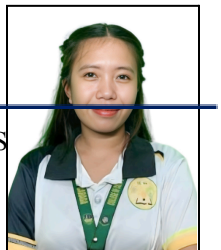
Educational Attainment

Elementary : Nasipit Central Elementary School
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Future Librarians and Information Scientists' Society (S.Y 2025-2026)	Public Information Officer
Future Librarians and Information Scientists' Society (S.Y 2024-2025)	Business Manager

CURRICULUM VITAE



NAME: RENALYN M. BENIGA
ADDRESS: Brgy
EMAIL: renalyn_beniga@smccnasipit.edu.ph
Personal Information

Sex	:	Female
Date of birth	:	February 11, 2005
Height	:	153 cm
Weight	:	48
Civil Status	:	Single

Educational Attainment

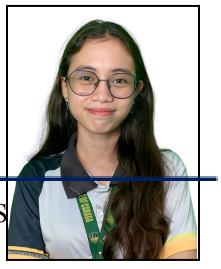
Elementary : Nasipit Central Elementary School
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Future Librarians and Information Scientists' Society (S.Y 2025-2026)	Third-Year Level Representative
Student Assistant Officer (S.Y 2025-2026)	Vice-President

CURRICULUM VITAE

NAME: CHERRISH JANE T. RANARIO
ADDRESS: Brgy



EMAIL: cherrishjane_ranario@smccnasipit.edu.ph

Personal Information

Sex	:	Female
Date of birth	:	June 12, 2005
Height	:	150 cm
Weight	:	40
Civil Status	:	Single

Educational Attainment

Elementary : Culit Elementary School
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Future Librarians and Information Scientists' Society (S.Y 2025-2026)	Peace Officer

CURRICULUM VITAE

NAME: KEVIN DOMINIC M. SUAN
ADDRESS: Amontay, Nasipit Agusan del Norte



EMAIL: kevindominic_suan@smccnasipit.edu.ph

Personal Information

Sex	:	Male
Date of birth	:	November 19, 2003
Height	:	157 cm
Weight	:	42.5
Civil Status	:	Single

Educational Attainment

Elementary	: Amontay, Elementary School
Secondary	: Carmen National High School
Tertiary	: Saint Michael College of Caraga

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

FSUU

SAINT MICHAEL COLLEGE OF CARAGA
A Tupon St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932
College of Computing and Information Sciences

October 20, 2025

Participation is still voluntary.

FATHER SATURNINO URIOS UNIVERSITY
BUTUAN CITY, PHILIPPINES
APPROVED
DATE: 10/20/2025 TIME:
DR. ARLYN M. FLORETA
VICE PRESIDENT
ACADEMIC AFFAIRS AND RESEARCH

DR. ARLYN M. FLORETA
VP for Academic Affairs & Research
Father Saturnino Urios University
Butuan City, Philippines 8600

Dear Dr. Floreta,

I trust this letter reaches you in good health.

I am writing to respectfully request permission to conduct our research study entitled **"Library Modernization: Influence in Students' Library Experience in Selected Academic Libraries in Agusan del Norte"** at Father Saturnino Urios University Library as part of our academic requirements.

The purpose of this study is to examine how modernized library facilities and services influence the overall library experience of students. Data will be collected through surveys administered to individuals present in the library during the conduct of the study.

All information gathered will be treated with the utmost confidentiality and used solely for academic purposes

Your favorable consideration of this request will be greatly appreciated. Should you have any questions or require further clarification, I may be reached through angelmicagonzales4@gmail.com or at 09359183032.

Thank you very much for your time and kind assistance.

FATHER SATURNINO URIOS UNIVERSITY
OFFICE OF THE VICE PRESIDENT FOR ACADEMIC AFFAIRS & RESEARCH
BUTUAN CITY, PHILIPPINES
RECEIVED
MAIL: ____ E-MAIL: ✓ HAND OVER: ____
BY: [illegible] CONTROL NO.: 25-26-0275
DATE: 10-20-25 TIME: 11:45am NO. OF PAGES: 2

Respectfully yours,

ANGELMICA J. GONZALES
Group Leader

Member

SMCC

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932
College of Computing and Information Sciences

October 17, 2025

REV. FR. RONNIEL G. BABANO, STL
School President
Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan del Norte, 8602

Dear Rev. Fr. Babano,

I trust this letter reaches you in good health.

I am writing to respectfully request permission to conduct our research study entitled **"Library Modernization: Influence in Students' Library Experience in Selected Academic Libraries in Agusan del Norte"** within the premises of Saint Michael College of Caraga Library as part of our academic requirements.

The purpose of this study is to examine how modernized library facilities and services influence the overall library experience of students. Data will be collected through surveys administered to individuals present in the library during the conduct of the study.

All information gathered will be treated with the utmost confidentiality and used solely for academic purposes

Your favorable consideration of this request will be greatly appreciated. Should you have any questions or require further clarification, I may be reached through angelmica_gonzales@smccnasipit.edu.ph.

Thank you very much for your time and kind assistance.

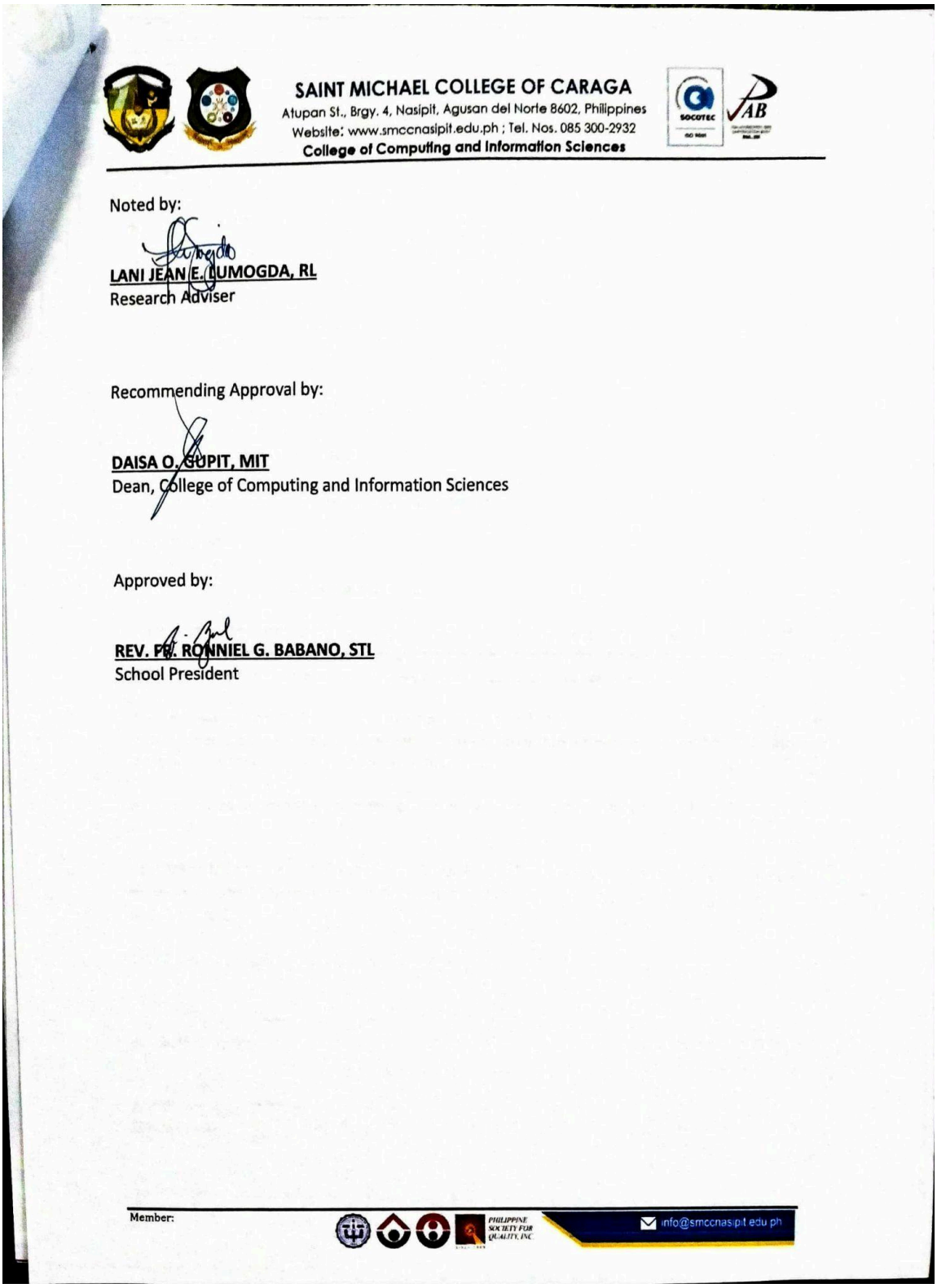
Respectfully yours,

ANGEL MICA J. GONZALES
Group Leader

SAINT MICHAEL COLLEGE OF CARAGA
NASIPIT, AGUSAN DEL NORTE
OFFICE OF OSP
RECEIVED
DATE: 10-23-25 CONTROL #:
BY: TIME: 8:18 am

Member:

info@smccnasipit.edu.ph



SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932
College of Computing and Information Sciences

Noted by:

LANI JEAN E. LUMOGDA, RL
Research Adviser

Recommending Approval by:

DAISA O. GUPIT, MIT
Dean, College of Computing and Information Sciences

Approved by:

REV. FR. RONNIEL G. BABANO, STL
School President

Member:

PHILIPPINE SOCIETY FOR QUALITY, INC.

info@smccnasipit.edu.ph

CSU

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan Del Norte, Philippines
Tel. Nos +63 085 363-3251 / +63 085 283-3113 Fax No. +63 085 908-0992
www.smccnasipit.edu.ph

October 27 , 2024

Dr. Rolyn C. Daguil
University President
Caraga State University
Ampayon, Butuan City

Subject : *Endorsement Letter for Data Gathering*

Dear Dr. Daguil,

Good Day!

I am writing this letter to formally endorse the research group composed of Angel Mica J. Gonzales, group leader, whose members are Ashlly J. Llavan, Jane Ranario, Kevin Dominic Suan and Renalyn Beniga, bona fide students of Bachelor of Library and Information Science (BLIS) under the College of Computing and Information Science (CCIS) at Saint Michael College of Caraga.

The group is currently conducting a research study entitled ***"Library Modernisation: Influence in Students' Library Experience in Selected Academic Libraries in Agusan del Norte"*** as part of our academic requirements.

In line with this, we respectfully request permission to gather general information about your academic library, which will serve as the baseline for identifying and selecting the research locales for our study. To accomplish this, we have attached a *Library Information Form* for your kind completion.

The information collected will be used solely for research purposes and will remain confidential.

We sincerely appreciate your time and cooperation in supporting our research.

Member:
info@smccnasipit.edu.ph

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan Del Norte, Philippines
Tel. Nos +63 085 363-3251 / +63 085 283-3113 Fax No. +63 085 908-0992
www.smccnasipit.edu.ph

Should you have any questions or require further clarification, you may contact me at angelmica_gonzales@smccnasipit.edu.ph.

Thank you very much for your kind assistance and valuable support.

Respectfully yours

LANI JEAN E. LUMOGDA, RL
Research Adviser

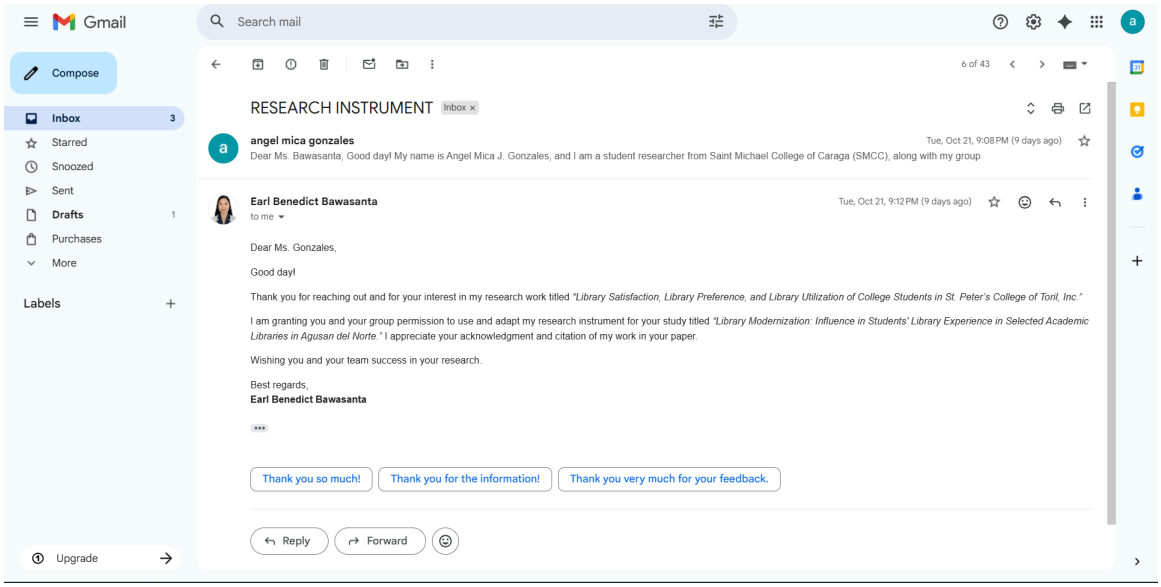
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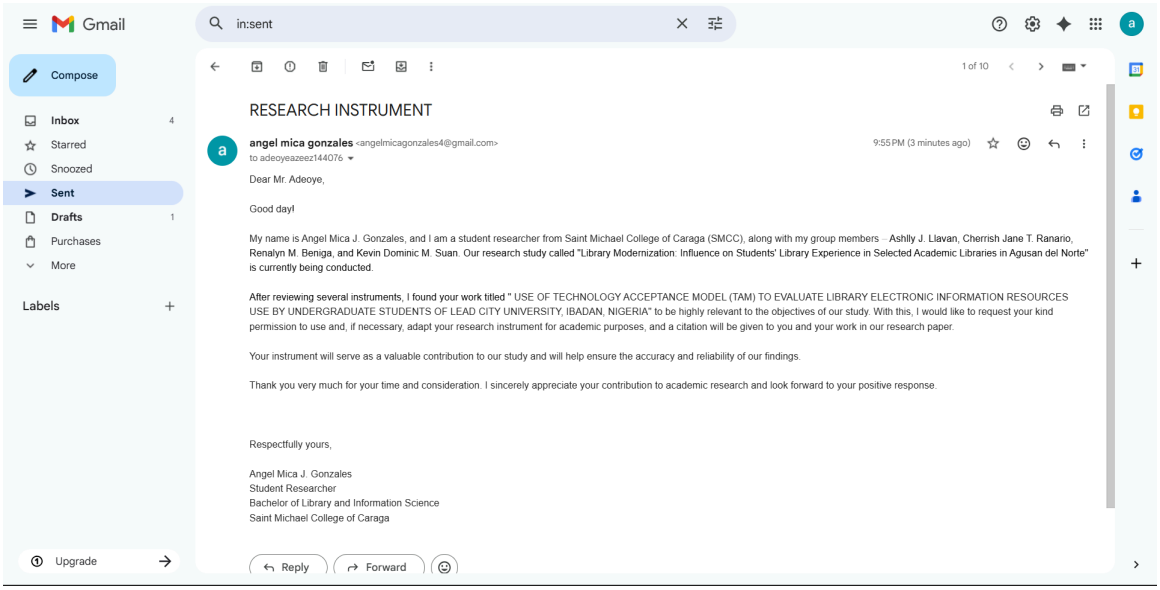
DAISA O. GUPIT, MIT
Dean, College of Computing and Information Sciences

Approved by:

DR. KENNETH IAN BARRERA
Head, Research and Instructional Innovation

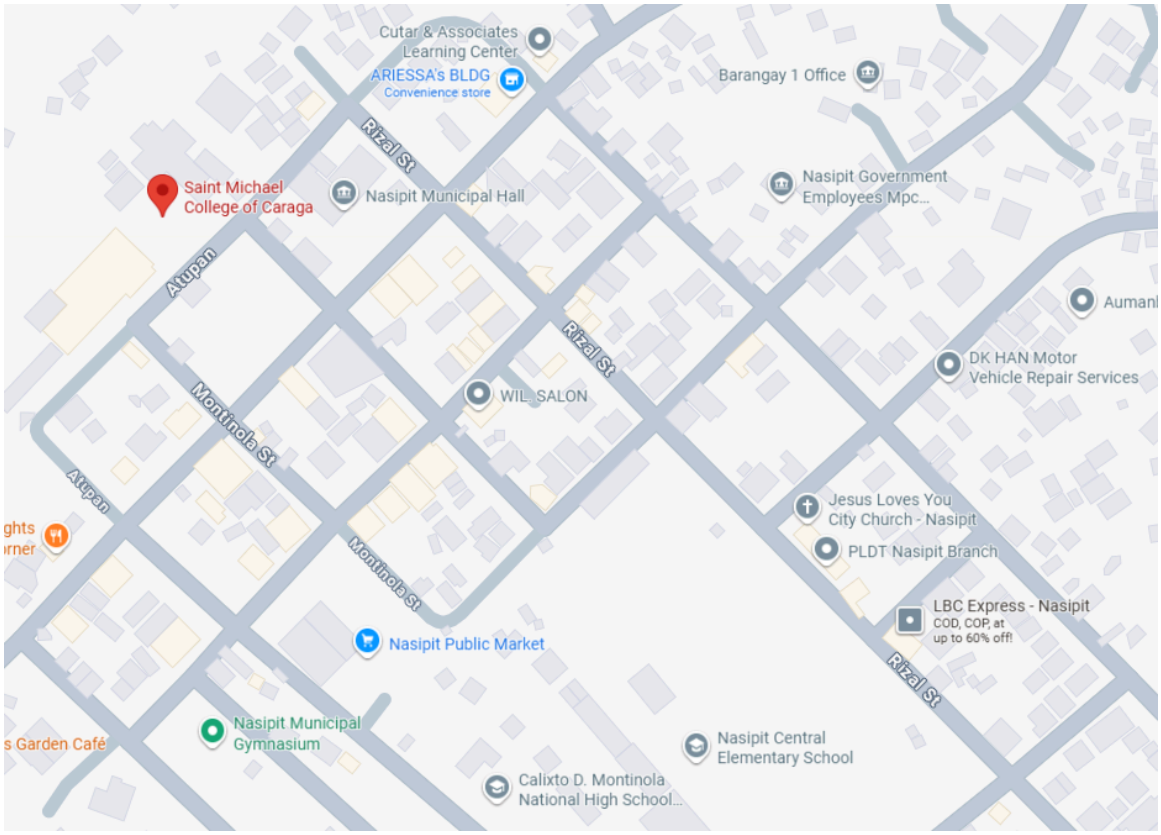
B - Instrument with Informed Consent



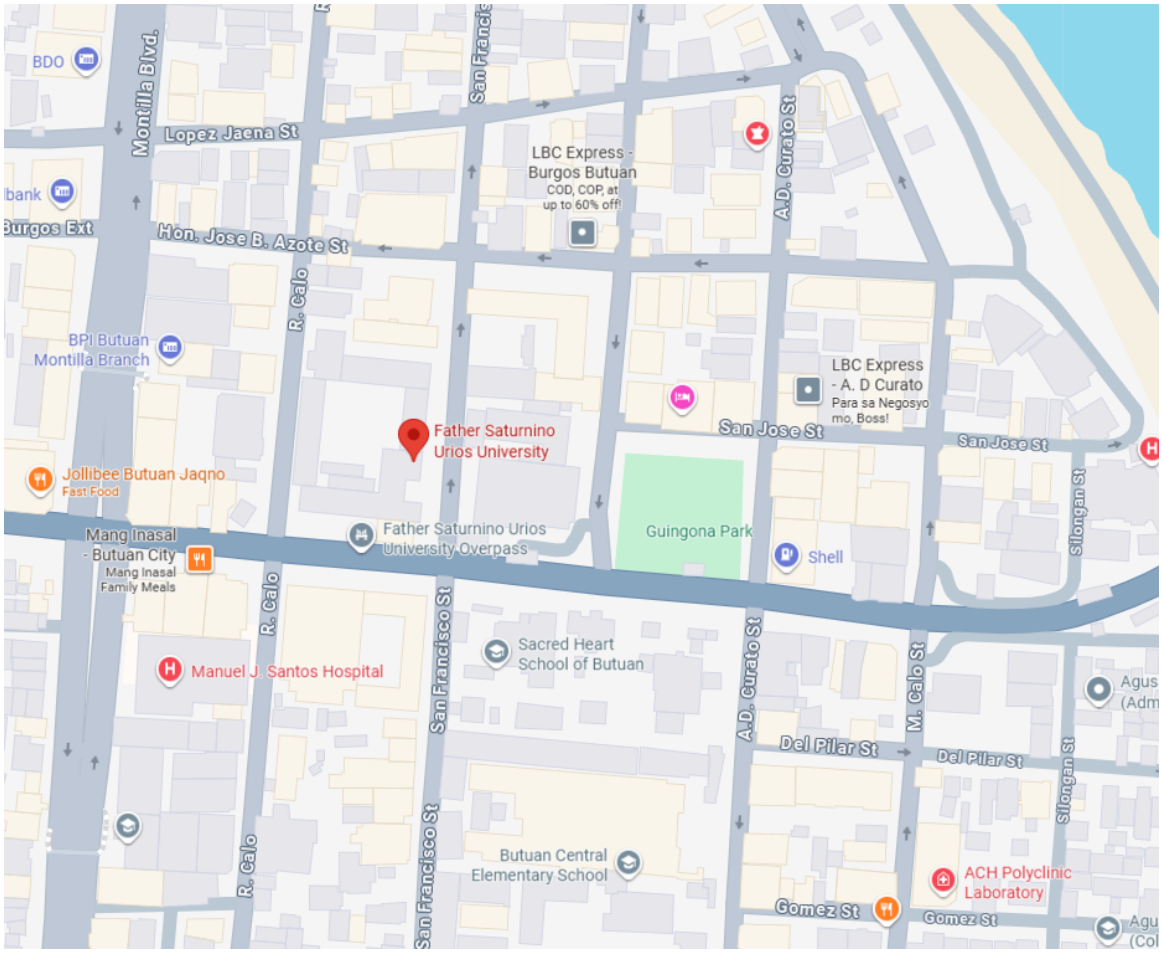


C - Map of the Research Locale

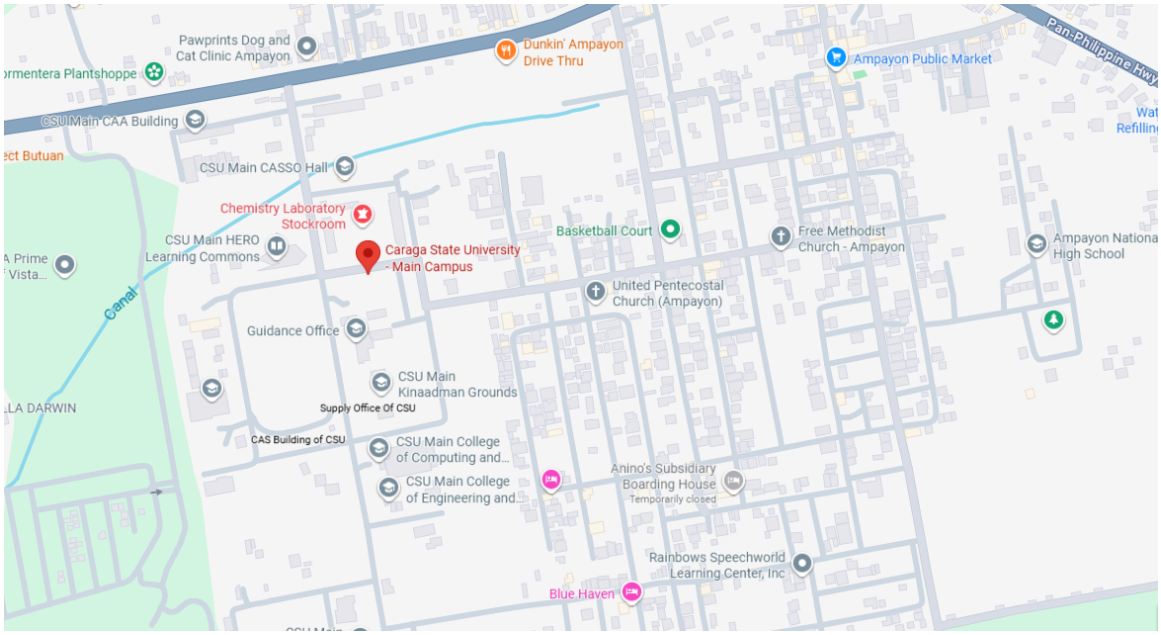
SAINT MICHAEL COLLEGE OF CARAGA



FATHER SATURNINO URIOS UNIVERSITY



CARAGA STATE UNIVERSITY



D - Narrative & Photo Documentation

Title Defense



Formulating of Chapter 1



Formulating Chapter 2



E- Ethics Committee Approval

CERTIFICATE OF TECHNOLOGY ASSESSMENT

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

TEST RESULT

TECHNOLOGY-BASED QUALITY ASSURANCE

Date of Test	October 29, 2025	**Control number**	**RBLIS125001**	
Research Title	**LIBRARY MODERNIZATION: INFLUENCE ON STUDENTS' LIBRARY EXPERIENCE IN SELECTED ACADEMIC LIBRARIES IN AGUSAN DEL NORTE**			
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Test subject	Chapters 1-2				
Total number of words	**6,870**	**Sentences**	**391**	**Characters**	**46,416**

	Test Area	Score	Required	Remarks
	(1) Plagiarism	**98**	**90%**	PASSED
	(2) Grammar	**97**	**95%**	PASSED
	(3) Readability Score	**32**	**25%**	PASSED
	(4) AI Text Detection	**26**	**40%**	PASSED
	(5) Reliability *(Test Instrument only)*	N/A	N/A	

FINDING(S) & RECOMMENDATION(S)

Evaluated by:	Approved by:
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SURVEY QUESTIONNAIRE

Title of the Study:

"Library Modernization: Influence in Students' Library Experience in Selected Academic Libraries in Agusan Del Norte"

INSTRUCTIONS:

Please read each item carefully and put a check mark (✔) on the option that best represents your answer. Your responses will be kept confidential and used for academic purposes only.

SCALE:

5 — Very Satisfied
4 — Satisfied
3 — Neutral
2 —Dissatisfied
1 — Very Dissatisfied

Name (Optional): _____

Date: _____

Part I. RESPONDENT PROFILE

Please check ✔ or fill in the appropriate information.

1. **Age**

   ☐ 18–21 ☐ 22–25 ☐ 26 and above

2. **School**

   ☐ FSUU ☐ CSU ☐ SMCC

3. **Year Level**

   ☐ 1st Year ☐ 2nd Year ☐ 3rd Year ☐ 4th Year

4. **Frequency of Visiting Libraries**

   ☐ Daily ☐ Monthly

   ☐ Weekly ☐ Rarely

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Part II. LIBRARY MODERNIZATION

Check ✔ that best describes the level of satisfaction

Digital Infrastructure	5	4	3	2	1
Wi-Fi connection within the library is stable and fast					
Sufficient supply of power outlets					
The computer workstations are available and user-friendly					
I can easily access technological tools and outlets for my gadgets					
The technical support for problems with the electronic system is satisfactory.					
The hardware (e.g., computers) available for accessing Learning Electronic Information Resources is sufficient and up-to-date.					
Overall, I am satisfied with the library's digital infrastructure for accessing electronic resources.					
Availability of Online Resources	**5**	**4**	**3**	**2**	**1**
Accessibility of updated references and online databases.					
The digital catalog provides useful and current resources.					
The library databases and e-resources are reliable, accessible, and updated.					
The variety of electronic journals and databases available through the library meets my course-related needs.					
I am satisfied with the relevance and currentness of the electronic books available for my subjects.					
I am aware of the full range of available Library Electronic Information Resources.					
The range of E-Journals and Databases available through the library meets my course-related needs.					
Overall, the availability of specific online resources (E-Journals, E-Books, etc.) is satisfactory for my academic tasks.					
Automation of Library Services	**5**	**4**	**3**	**2**	**1**
The borrowing, returning, renewal, and preservation process is efficient.					

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Printing, photocopying, and scanning services are available and easy to use					
Online Public Access Catalog (OPAC) is easy to use and accessible.					
The modernized systems make my library experience more convenient					
I am satisfied with the ability to access subscribed resources (e.g., databases) remotely (off-campus).					
The electronic system provides helpful guidance/feedback when I perform tasks					
Overall, the automation of library services (e.g., search, access) makes my academic work easier.					
Enhanced Physical Learning Process	**5**	**4**	**3**	**2**	**1**
The library environment encourages to stay longer and study.					
Rooms for group discussions and collaborative study are adequate.					
There are sufficient study areas and reading spaces for all users.					
Availability of information and circulation desks.					
The library is well-ventilated and maintains good air quality.					
The library design accommodates students with special needs.					
The spaces are flexible and can be used for different learning needs					
Comfortable lighting, temperature, and furniture design.					
I can select areas with appropriate noise levels based on my study needs.					
I can adjust some furnitures (chairs, tables) and lighting based on my comfort and preferences.					
The library space encourages collaborative learning activities when using LEIR.					
Overall, I am satisfied with how the physical spaces support my use of Library Electronic Information Resources.					

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Part III. STUDENTS' LIBRARY EXPERIENCE

Check ✔ that best describes the level of satisfaction

Ease of Access to Library Service	5	4	3	2	1
It is easy to locate materials and services I need.					
The arrangement of books and materials is logical and organized.					
The library services are designed to be accessible to all users					
I find it easy to recover from errors encountered while using Learning Electronic Information Resources.					
It is easy for me to remember how to perform tasks using the Learning Electronic Information Resources.					
Interacting with the Learning Electronic Information Resources system does not require a lot of my mental effort.					
Overall, I find the Library Electronic Information Resources easy to access and use.					
Perceived Usefulness of Modernized Features	**5**	**4**	**3**	**2**	**1**
The library resources are useful for my research and academic tasks.					
The library supports my learning productivity and outcomes.					
Using Learning Electronic Information Resources enhance my effectiveness on my academic work.					
Using Learning Electronic Information Resources (E-Journals, E-Books) gives me greater control over my course work.					
Using Learning Electronic Information Resources enhance my effectiveness on my academic work.					
Overall, I perceive the Library Electronic Information Resources as highly useful for my academic tasks.					
User Satisfaction	**5**	**4**	**3**	**2**	**1**
I am satisfied with the overall modernized features and services.					
The library staff are helpful, approachable, and responsive.					
The library provides prompt and efficient assistance.					

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I am frequently happy when using Library Electronic Information Resources.					
I am satisfied with the level of training or assistance provided by the library staff					
Overall, I am satisfied with my experience with the Library Electronic Information Resources.					
Perceived Impact on Academic Productivity	**5**	**4**	**3**	**2**	**1**
The available information helps me improve my study habits and academic performance.					
The library materials contribute to my academic growth and self-study.					
The library plays an important role in enhancing my knowledge and skills.					
Using Learning Electronic Information Resources saves me time in completing my academic assignments.					
Library Learning Electronic Information Resources and Services enable me to accomplish tasks more quickly.					
Library learning resources and services improves the quality of the academic work I do.					
Library learning resources and services Increases my productivity in research and study.					
Overall, the Learning Electronic Information Resources and Services have a positive impact on my academic productivity.					

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INFORMED CONSENT FORM

Title of the Study:
"Library Modernization: Influence in Students' Library Experience in Selected Academic Libraries in Agusan Del Norte"

Dear Respondent,

We, the researchers, conducted a study entitled ***"Library Modernization: Influence in Students' Library Experience in Selected Academic Library in Agusan Del Norte"*** as part of our academic requirements for Bachelor of Library and Information Science program. The purpose of this research is to examine how modernized library facilities and services influence the overall library experience of students.

If you agree to take part, your participation will involve answering a short survey questionnaire. This will take about 10–15 minutes of your time. There are no right or wrong answers and your honest responses are valuable for the success of this study. Your participation is completely voluntary. You are free to decline or withdraw from this study without any consequences.

All the information you provide will be kept strictly confidential. Your name will not appear in any report or publication resulting from this study. The data collected will be used solely for research purposes and will be stored securely by the researcher.

CONSENT STATEMENT
By checking the box below or signing your name, you are indicating that:

- You have read and understood the information above.
- You voluntarily agree to participate in this study.
- You are at least 18 years old and above.

☐ I have read and understood the information provided, and I voluntarily agree to participate in this study.

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Participant's Name (optional): _____

Signature: _____

Date: _____

CERTIFICATE OF AUTHENTIC AUTHORSHIP

This is to certify that I/we, the undersigned author(s), have solely and honestly written the entire paper titled: **"Library Modernization: Influence on Students' Library Experience in Selected Academic Libraries in Agusan del Norte"** All contents, ideas, data, and analysis presented in this paper are the results of my/our own diligent work and intellectual effort. The paper has been crafted to the best of my/our abilities, in full compliance with academic integrity standards and ethical research practices.

I/We further declare that:

1. This paper is free from any form of plagiarism or copyright infringement.
2. All referenced materials have been properly cited, and due credit has been given to the original authors and sources.
3. The work does not contain any falsified or fabricated data or information.

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Signed this

Date: Click here to enter text.

Location: Click here to enter text.

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